

Analyzing Secrecy Performance in Wireless Communication Networks with Randomly Distributed Eavesdroppers

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Abstract—Emerging wireless communication technologies, including Ultra-Wideband (UWB), 5G New Radio (NR), and Long-Range Wide Area Networks (LoRaWAN), provide high data rates and expansive coverage, yet their broadcast nature makes them vulnerable to unauthorized interception. Existing physical-layer security (PLS) models typically evaluate dual-hop transmission under static eavesdropper assumptions or single-hop random eavesdropping. To address these limitations, this paper proposes a generalized secrecy performance analysis for wireless networks subject to randomly distributed eavesdroppers over $\alpha-\mu$ generalized fading channels in a dual-hop Single-Input Multiple-Output (SIMO) configuration. We investigate three realistic operational scenarios: (i) eavesdroppers intercepting the source link, (ii) eavesdroppers intercepting the relay link, and (iii) eavesdroppers simultaneously attacking both the source and relay hops. Closed-form analytical expressions are derived for key secrecy performance metrics, including the secrecy outage probability (SOP), strictly positive secrecy capacity (SPSC), intercept probability (IP), and effective secrecy throughput (EST). The derived expressions capture the intricate impacts of channel fading severity, multi-antenna reception diversity, and spatial eavesdropper density on secrecy performance.

Index Terms—Secure outage probability (SOP), strictly positive secrecy capacity (SPSC), intercept probability (IP), effective secrecy throughput (EST), $\alpha-\mu$ fading, decode-and-forward (DF) relay, physical layer security (PLS), SIMO.

I. INTRODUCTION

A. Background and Related Works:

In recent years, there has been a great interest in wireless channel secrecy research due to the growing importance of wireless communication, particularly in the context of 5G and beyond. Keeping wireless networks secure has grown crucial as they continue to support a wide range of applications. Secrecy analysis has become a key component in the pursuit of robust and secure wireless communication because of the additional obstacles the impending quantum computing era has brought forth. These challenges have motivated academics to investigate novel strategies to safeguard wireless channels against new threats.

In regards to the most recent developments in wireless communication technology, researchers are both optimistic and wary. On the one hand, they are aware of the enormous promise of technologies like 5G, Free Space Optics, and others

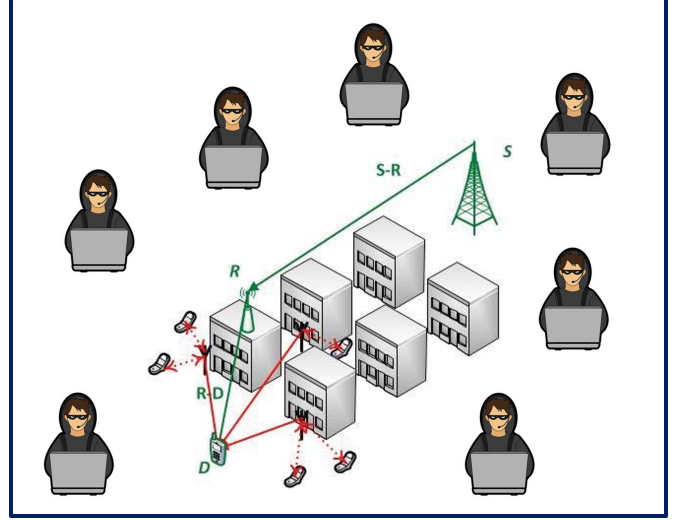


Fig. 1. Randomly Distributed Eavesdropper

in terms of transforming connectivity and enabling cutting-edge applications. Researchers are highly aware of these innovations' security challenges, though. They stress that as these technologies get more intricate and intertwined, the attack surface widens, leaving them open to a variety of dangers such as eavesdropping, data breaches, and network outages. Researchers are continuously investigating advanced security protocols, intrusion detection systems, and security measures to allay these worries and make sure that the advantages of these cutting-edge wireless technologies may be reaped without jeopardizing security and privacy.

1) *Security using dual hop:* In a dual-hop dual-relay (DH-DR) network [1], where two relays help communication between the source and destination without a direct link, the study looks at the physical-layer security using traditional approaches. An ordinary modulation method like 4-QAM is used in the standard DH-DR arrangement. The paper does, however, provide a signal-space approach that, by having the relays convey various symbols to the destination, considerably improves security. A significant 4 dB improvement in secrecy outage probability is shown by the analysis, which primarily focuses on intercept probability and secrecy outage probability. Additionally, the suggested system is contrasted with the current system to demonstrate its superiority. Monte Carlo simulations provide additional evidence of the proposed system's improved performance.

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The performance of a dual-hop hybrid system that combines RF and VLC for visible light communication is examined in this research [2] in terms of secrecy. In this configuration, a source (S) sends data over RF to a relay (R), which decodes the signal over RF and sends it over VLC to a destination (D). The system makes the assumption that there is an observer (E) trying to intercept the RF signal while it is being transmitted. A specific distribution is used to characterize the RF and VLC channels, and a random waypoint model is used to describe the receiver's mobility. Meijer's G-function is used in the paper to construct analytical equations for average secrecy capacity and intercept probability, giving researchers a tool to evaluate how changes to various channel parameters, such as RF shadowing, VLC field of view, and Lambertian emission, affect secrecy performance.

In this paper [3], they investigate hybrid satellite-terrestrial (HST) networks as potential replacements for conventional cellular systems in the context of the high data rate requirements of the 5G future. This study focuses on wireless multicasting in a HST relaying network (HSTRN), where a satellite distributes data to terrestrial nodes across numerous relays while potentially encountering eavesdroppers. Opportunistic relaying is used to secure multicast transmissions. Alpha-mu fading occurs on terrestrial networks while Shadowed Rician fading occurs on satellite links. The study offers closed-form formulas for important metrics like secure outage probability, ergodic secrecy multicast capacity, and nonzero secrecy multicast capacity probability. These tools expose secrecy trade-offs including relay count, multicast users, and eavesdroppers, as well as ways for maintaining perfect secrecy in difficult channel conditions.

The performance and security of cutting-edge wireless communication technologies like 5G and 6G are being actively improved by researchers [4]. Reconfigurable Intelligent Surface (RIS), which improves electromagnetic wave propagation conditions for greater efficiency, is one possible method. Additionally, UOWC (Underwater Optical Wireless Communication) is replacing conventional acoustic and RF technologies in underwater communication. This study focuses on the mixed dual-hop RIS-assisted RF-UOWC networks' secrecy aspect while taking the threat of eavesdropping into account. While the UOWC link works with a mixture of exponential generalized Gamma distribution, the RF link undergoes generalized Gamma fading. The study presents analytical formulas for average secrecy capacity, secrecy outage probability, and other crucial metrics, accounting for elements like turbulence, temperature gradients, and salinity gradients, making it especially pertinent for temperature gradients.

Underwater high-speed optical wireless communication, which can supplement or replace conventional RF and acoustic technologies, is the subject of research [5]. This study examines how well a hybrid dual-hop RF-UOWC system with variable gain relays performs in terms of secrecy when exposed to possible eavesdropping. Generalized Gamma fading occurs on the RF link while Exponential Generalized Gamma distribution is encountered on the UOWC link. For positive secrecy capacity, average secrecy capacity, and secrecy outage probability, the secrecy analysis produces closed-

form formulations. Unlike earlier models that concentrated on thermally homogenous settings, this model takes into account temperature gradient scenarios and air bubbles in RF-UOWC. The formulas that were derived are verified by Monte-Carlo simulations.

This study [6] examines the physical layer security (PLS) of a dual-hop mixed RF/FSO communication system, which consists of both RF and optical lines, in the presence of listeners. When taking into account an urban setting, the RF linkages exhibit Rayleigh distribution whereas the FSO links exhibit Málaga distribution. Effective secrecy throughput (EST) and secrecy outage probability (SOP), for example, are closed-form metrics for secrecy performance that the study develops. The effects of diversity approaches like selection combining (SC) and maximal ratio combining (MRC) are also examined, as well as the practical features of FSO links including atmospheric turbulence, optical signal detecting methods, and pointing problems. Asymptotic expressions are derived, and the validity of the results is checked using mixed FSO/RF systems and Monte-Carlo simulations.

The use of friendly jammers to increase information security against eavesdroppers is the main focus of this paper's [7] investigation into physical layer security (PLS) in wireless communication. The article suggests a paradigm for resource optimization for a dual-hop relay network with the goal of increasing the sum secrecy rate through effective subcarrier assignment and appropriate power distribution among subcarriers at base stations and relay nodes. To identify effective solutions, they use Lagrangian dual methods and evaluate their strategy against three benchmark frameworks. Results from simulations show how their joint resource optimization paradigm is superior for obtaining improved PLS in wireless communication, especially when cooperative jamming is involved.

This study [8] examines secure satellite communication, which is important for many applications. It addresses security issues by introducing a dual-hop communication system that uses a satellite as an amplified-and-forward relay. The work gives theoretical models for connection failure probability and secrecy, examines security-reliability tradeoffs, and suggests transmit power strategies based on search algorithms. The power schemes are made simpler by the development of asymptotic formulas for outage performance. The suggested secure transmission model is validated through simulations, which also show that transmit power methods, led by asymptotic analysis, efficiently enhance the performance of relay satellite communication systems while boosting security in wireless networks.

This study [9] investigates the secrecy performance of a dual-hop UAV-assisted relay network, which provides mobility and cost-effectiveness in comparison to ground-based relaying techniques. The analysis uses a stochastic geometric approach to predict how the ground user and eavesdroppers are distributed spatially. Cumulative distribution functions (CDF) for the user and eavesdroppers are found using this paradigm. The study takes into account a decode-and-forward (DF) relay protocol and produces analytical formulas for strictly positive secrecy capacity (SPSC), average secrecy capacity

(ASC), and network security outage probability (SOP). The analytical results are validated by Monte-Carlo simulations, which further illustrate the effects of various network and channel factors on secrecy performance, including fading, eavesdropper density, SNR, UAV height, and coverage radius.

This study [10] investigates the security of a mobile, dual-hop UAV-based wireless communication system with imprecise channel state information. Two eavesdroppers attempt to intercept communications through a decode-and-forward relay that connects a source and a destination in the system. A closed-form formula is derived by the study using statistical characteristics of the signal-to-noise ratio for the network's secrecy intercept probability. Based on the findings, secrecy is improved by raising the average SNR and the number of antennas at legal receivers. But secrecy deteriorates as CSI imperfection, node speed, delay, and carrier frequency increase. The results are supported by Monte Carlo simulations.

In a dual-hop multi-secondary user (SU) cognitive radio network, this research [11] investigates the performance of the intercept probability (IP). The simultaneous wireless information and power transmission (SWIPT) protocol is used to power the relay autonomously by energy harvesting from secondary sources. While dealing with a primary network and eavesdroppers at each hop, single-antenna SUs convey data to a multi-antenna end-user through the relay. The analysis takes into account variables including transmit power, degree of fading, and maximum allowable interference power. In order to account for the correlation between the secrecy capacities for the two hops, a closed-form formula for IP is developed. In particular, in low interference and transmit power conditions, Monte Carlo simulations corroborate the findings and shed light on the security of the system.

The performance of a dual-hop RF-UOWC (radio frequency-underwater optical wireless communication) system's secrecy is examined in this paper [12]. Eavesdroppers are modeled using stochastic geometry, and for RF and UOWC links, respectively, Nakagami-m and mixed exponential-generalized gamma distributions are used. In different settings and with different relaying mechanisms, the study establishes statistical rules for signal-to-noise ratios. The lower bound of the secrecy outage probability (SOP) is expressed in closed-form. The analytical results are validated by Monte Carlo simulations, which also show the influence of ambient variables on the SOP of RF-UOWC systems, such as temperature and bubble levels.

2) *Security on randomly eavesdropper*: This work [13] presents a secure communication method for content-based image transmission that is learning-aided. It employs a hybrid strategy to address the variable significance of various image regions for security. While secure confidential packets are inserted with artificial noise, public regions are transmitted error-free. Security is improved via fountain-based packet delivery, and transmission parameters are optimized by a deep neural network. A closed-form quality-of-security (QoSec) violation probability expression is used to quantify the secrecy performance. The results of the simulations show that this learning-assisted method performs better than the alternatives, meeting the security and latency requirements to a significant

extent.

The issues encountered by eavesdroppers within the narrow beam are addressed by the deep neural network (DNN)-based solution for secure Terahertz (THz) communications presented in this study [14]. The suggested approach takes into account randomly placed eavesdroppers and uses artificial noise (AN) to preserve self-interference cancellation (SIC). To reduce the likelihood of a secrecy outage, the system parameters are improved with a DNN method by utilizing the temporal widening effects of AN signals at various distances. Results reveal a maximum secrecy rate of 3.3 bps/Hz over 10 meters, and compared to traditional transmitter AN schemes, the risk of a secrecy outage is substantially lower. This makes it appropriate for THz secure communication in the presence of listeners.

This article [15] examines a low-latency, ultra-reliable communication system with brief packets and no feedback loop to determine its secure spectral efficiency. With a Nakagami-m fading channel and multiple cooperating eavesdroppers in mind, it contrasts the performance-enhancing effects of packet replication (PR) and interface diversity (ID) techniques for physical-layer security. The results show that to maximize safe spectral efficiency, it is crucial to jointly optimize blocklength, transmit power, and information bits per codeword. In most instances, ID also outperforms PR as the number of replications or interfaces increases and provides insights into safeguarding communication under strict latency limitations.

In downlink transmissions with active eavesdropper placed at random, the physical layer security issue is addressed in this letter. To safeguard the sensitive signal, artificial noise (AN) is introduced. It is suggested [16] to use a hybrid outage probability metric to assess security performance that takes into account both transmission and secrecy outage possibilities. This statistic is the foundation for power allocation optimization, and an insecure area is set up to limit where an eavesdropper can go. The receiver's immediate environment is insecure, as shown by the simulation findings.

The security and dependability of systems for ultra-reliable low-latency communication (URLLC) with randomly placed eavesdroppers are examined in this work [17], taking into account short blocklength codes and inaccurate channel estimate. It offers asymptotic formulations for intercept probability and closed-form approximations for transmission error probability while taking secrecy protected zones into consideration. The findings are supported by numerical and simulation studies, which further highlight the trade-off between security and dependability. For secure URLLC applications in upcoming 6G wireless networks, the paper gives suggestions and design principles.

This paper [18] examines the privacy performance of a re-configurable intelligent surface (RIS)-assisted communication system with unmanned aerial vehicle (UAV) eavesdroppers that are dispersed at random. It takes into account scenarios with one or more antennas at the base station (BS) and derives analytical equations for the signal-to-noise ratios (SNRs) of authorized users and eavesdroppers. In closed-form expressions, ergodic secrecy capacity is also roughly calculated. The accuracy of the analytical and approximative formulations is

confirmed by the simulation results, which also show how the use of RIS increases security.

This article [19] focuses on passive eavesdroppers and an untrusted relay when discussing physical layer security (PLS) in millimeter-wave (mmWave) communication systems. To increase the instantaneous secrecy rate, cooperative jamming between the source and destination is recommended for network protection. The paper suggests a new solution for the non-colluding eavesdroppers (NCE) optimal power allocation (OPA) problem and produces a formula for the ergodic secrecy rate (ESR) over mmWave fading channels, accounting for blockage with a model that includes both line-of-sight (LoS) and non-line-of-sight (NLoS) pathways. The proposed OPA strategy has better ESR according to simulation findings than previous plans.

This study [20] examines security issues in wireless sensor networks with both passive and active eavesdroppers that are dispersed randomly. Active listeners have the ability to jam and eavesdrop simultaneously. The work suggests a defense technique that establishes a protected area utilizing beam-forming and artificial noise (AN). By taking into account both transmission and secrecy outage probabilities, it assesses security performance. To evaluate security, the idea of a safe transmission range is presented. The unsecured area and the effect of active eavesdroppers on wireless sensor network transmission are shown by the simulation findings.

In wireless healthcare systems, this research [21] examines a content-aware medical image transmission technique. It deals with the problem of secure transmission when there are passive listeners. A region of interest (RoI) and backdrop are separated from the overall image, each with its own set of security constraints. When transmitting RoI packets, intercept probability is regulated to reach specified thresholds. To improve security and maintain picture confidentiality while meeting latency constraints, the approach leverages fountain-based packet delivery. The outcomes of the simulation show that in a real-world medical imaging scenario, dependable file transfer and image secrecy are possible.

This study [22] investigates how random jitter in UAV platforms affects the performance of air-to-ground (A2G) wiretap systems' secrecy. It examines how UAV jitter affects the beam-illumination probability and illumination area of ground terminals and creates a framework for examining secrecy coverage probability (SCP) and ergodic secrecy capacity (ESC). When UAV height and beamwidth are properly set, the study finds that UAV jitter can improve secrecy performance in A2G wiretap systems—a novel conclusion that has been thoroughly proven by simulations.

This study [23] investigates the behavior of valid device-to-device (D2D) communication lines when they are intercepted by randomly dispersed eavesdroppers. To account for both large-scale and small-scale signal fading, the study introduces the idea of the k -th best eavesdropper. It focuses on calculating the likelihood of interception (P_{int}) and the chance of a secrecy outage (P_{out}). Based on the probability density function and cumulative distribution function of the received signal-to-noise ratio for the k -th best eavesdropper, the paper offers closed-form formulations for P_{int} and P_{out} using univariate

and bivariate Fox's H-functions. The paper investigates the effects of eavesdropper density, fading parameters, and path-loss exponent on the chance of interception and uses Monte-Carlo simulations to verify the analytical results. Notably, the genuine D2D link is seriously threatened by the first-best eavesdropper.

3) *Dual hop and simultaneous eavesdropping*: A single static eavesdropper who may attack in three cases like source, relay and both source-relay is considered in this paper [24]. They perform secrecy analysis on mixed $\eta - \mu$ fading channel and DGG cooperative relaying system. The researchers calculate the secrecy performance based on strictly positive secrecy capacity and secure outage probability. Utilizing these equations, they further examine how fading, turbulence, and pointing errors affect the secrecy performance of the (FSO-RF) system. Intensity modulation/direct detection (IM/DD) and heterodyne detection (HD) approaches are contrasted, with HD outperforming IM/DD in terms of secrecy performance. They found that calculation of scenario-3 where eavesdropper attack on both relay and source, becomes more difficult to predict. Additionally, a comparison of HD and IM/DD techniques shows that, when compared to the suggested system, the HD methodology delivers a better and more secure outage performance than the IM/DD technique.

In multi-cell massive MIMO systems, this work [25] investigates interactions between a base station and an active eavesdropper with a focus on blocking simultaneous eavesdropping and jamming. To reduce jamming, it introduces a beam extraction technique and a beam-domain reception mechanism. The usefulness of the suggested method in lowering the performance of active eavesdropping as compared to passive eavesdropping is demonstrated using a hierarchical game analysis without the usage of prior eavesdropper information. The direction and distance of the eavesdropper have an impact on the upper bound for listening. Finally, a proposal for an optimization procedure to increase system secrecy capacity is made.

This ground-breaking study [26] examines how reconfigurable intelligent surfaces (RIS) could improve multi-user cellular networks' spectrum efficiency and security. In order to maximize the worst-case sum rate while taking into account various constraints, it solves issues including incomplete angular channel state information. To address the complexity of the problem, the paper provides cutting-edge alternative optimization techniques, heuristic plans, and manageable formulations. It represents a considerable improvement in wireless network security by proving the higher performance and validity of the suggested solutions through numerical simulations.

In a hostile setting, it can be difficult to detect and stop tactical communications coming from intelligent transmitters. It uses a full-duplex (FD) approach to jam and eavesdrop at the same time, accelerating rate deterioration and data eavesdropping. An intelligence-enhanced FD jammer and its opponent play a power-domain Bayesian-Stackelberg game in the research [27], which aims to depict their adversarial interaction with little knowledge. The work establishes the existence of a Stackelberg equilibrium (SE) power solution and demonstrates the asymptotically optimality of the derived

semi-closed-form SE solutions. The suggested method performs better than existing benchmark techniques and greatly increases the usefulness of the jammer node.

The possibility of re-configurable intelligent surfaces (RIS) in improving the secrecy performance of a mixed radio frequency (RF) and free space optics (FSO) communication system is explored in this study [28]. The paper discusses scenarios involving simultaneous eavesdropping on both RF and FSO links, eavesdropping on FSO links, and RF-link eavesdropping. The investigation takes into account strategies for detection as well as Rician fading and Málaga-distributed turbulence with aiming errors. Secrecy metrics analytical formulas are offered, and the outcomes of Monte-Carlo simulations are used to validate the findings. The results highlight how important fading settings, air turbulence, and pointing errors are for enhancing secrecy performance.

This research [29] focuses on the security of a hybrid radio frequency (RF)-free-space optical (FSO) communication system featuring a decode-and-forward (DF) relay. The RF and FSO links are characterized by correlated $\eta - \mu$ and exponentiated Weibull fading distributions, respectively. The study addresses simultaneous eavesdropping by adversaries on both RF and FSO links. To enhance security, maximal ratio combining (MRC) and selection combining (SC) receivers are employed in the RF and FSO links, respectively. Closed-form formulas for secrecy outage probability and probability of non-zero secrecy capacity are derived using Meijer's-G functions. Numerical results and Monte-Carlo simulations illustrate the impact of various factors, including simultaneous eavesdropping and correlation, on the system's secrecy performance.

B. Motivation and contributions:

Recent years, researchers have proposed different models on wireless communication for secrecy analysis. Some of the papers are related to security analysis based on dual hop method considering one eavesdropper, some are security analysis based on randomly eavesdropper and remaining are based on simultaneous eavesdropping considering static eavesdropper position. But non of the paper considered secrecy analysis based on simultaneous eavesdropping randomly distributed eavesdropper based on latest dual hop method. This is the main contribution of our work. Our model can generalize all the previous model and solve the drawbacks of those models. Summary of our work are given below.

- 1) First, using the CDF and PDF of the signal-to-noise ratio (SNR) for each hop separately, we determine the cumulative distribution function (CDF) and probability density function (PDF) for the dual-hop system. Due to the fact that they take into consideration the particular generic fading traits displayed by our proposed $\alpha - \mu$ channel, these derived expressions are noteworthy since they are thought to be novel in comparison to other current systems.
- 2) By developing expressions for two secrecy performance metrics, SOP and SPSC, which are applicable in three different eavesdropping scenarios, we examine the secrecy performance of the system under consideration.

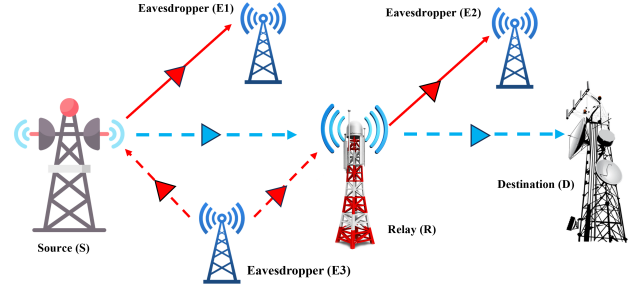


Fig. 2. System Model of Dual-Hop SIMO Relaying with Randomly Distributed Eavesdroppers.

Monte-Carlo (MC) simulations are used to further confirm the analytical results. Additionally, in order to make more insightful observations about the secure outage performance, we also derive the asymptotic SOP expression. Though the authors of [24] [23] suggested similar secure systems, the models proposed in this study have the ability to examine the secrecy performance under all turbulence situations, giving us an advantage over previous researches in our analysis. Additionally, we show that considering distance of randomly distributed eavesdropper, easily analyze their location and perform security.

- 3) We also calculate intercept probability (IP) and effective secrecy throughput (EST) to enhancing and assessing the security of wireless network in the scenarios with evolving vulnerabilities and threats. The probability of illegal data interception during transmission is measured by IP. It expresses the likelihood that a eavesdropper will be able to correctly decode the information being broadcast, jeopardizing the security of the system. As an alternative, EST is a performance metric that assesses the actual secure data rate attained when the attempts present of potential eavesdropping. In light of the built-in security features, it takes into account the system's capacity to uphold secrecy while maximizing the data throughput for authorized users.

C. Paper organization:

The paper is arranged in such a way that system model and problem formulation, scenario-I eavesdroppers attack in source, scenario-II eavesdroppers attack in relay and in scenario-III eavesdroppers attack in both source and relay. Rest of the part is Monte-Carlo simulations and remarks.

II. SYSTEM MODEL AND PROBLEM FORMULATION

As illustrated in Fig. 2, confidential data is transmitted from a single-antenna source (S) to a multi-antenna destination (D) via an intermediate decode-and-forward (DF) relay (R). The relay R is equipped with M receiving antennas and a single transmit antenna, while the destination D is equipped with M_1 receiving antennas. All legitimate and wiretap links experience independent and non-identically distributed $\alpha - \mu$ generalized fading. Non-colluding eavesdroppers (E_k), each

equipped with p antennas, are spatially distributed according to a homogeneous Poisson Point Process (PPP) with spatial density Z_e . Depending on the physical location and reception capability of the eavesdroppers, three operational wiretapping scenarios are investigated:

- **Scenario-I:** The eavesdroppers intercept confidential transmissions solely from the source ($S - R$ link). The second hop ($R - D$ link) remains secure.
- **Scenario-II:** The source link is secure, while eavesdroppers intercept transmissions emanating from the relay ($R - D$ link).
- **Scenario-III:** Both the source and relay hops undergo simultaneous and independent wiretap attacks by the distributed eavesdroppers.

III. SCENARIO-I

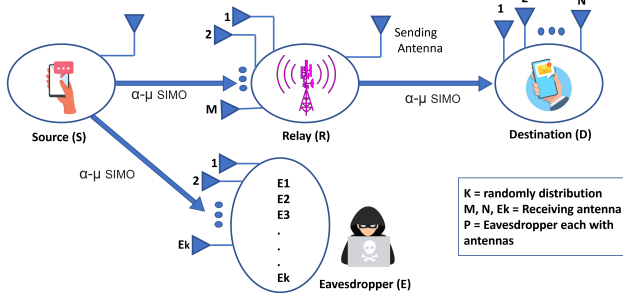


Fig. 3. Eavesdropping in the Source

A. Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of $S - R$ link

1) *PDF of $S - R$ link:* The PDF of the $S - R$ link can be given as [30],

$$f_{sr}(x) = \beta_1 x^{\beta_2} e^{-\beta_3 x^{\bar{\alpha}_r}}, \quad (1)$$

where $\beta_1 = \frac{\bar{\alpha}_r (M\mu_r)^{M\mu_r}}{\Gamma(M\mu_r)(M\bar{x}_r)^{M\mu_r\bar{\alpha}_r}}$, $\beta_2 = M\mu_r\bar{\alpha}_r - 1$, $\beta_3 = \frac{M\mu_r}{M\bar{\alpha}_r\bar{x}_r^{\bar{\alpha}_r}}$, α_r is the fading parameter of the $S - R$ link, M is the number of receiving antennas at the relay, μ_r represents the inverse of the normalized variance of $S - R$ Link, $\bar{x}_r$ is the average SNR of $S - R$ link, x is the instantaneous SNR of $S - R$ link, and $\bar{\alpha}_r = \frac{\alpha_r}{2}$.

2) *CDF of $S - R$ link:* The CDF of the the $S - R$ link can be obtained as [4],

$$F_{sr}(x) = \int_0^x f_r(x) dx, \quad (2)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$F_{sr}(x) = \beta_6 - \sum_{k_1=0}^{\beta_5-1} \beta_{55} x^{\beta_{54}} e^{-\beta_3 x^{\bar{\alpha}_r}}, \quad (3)$$

where $\beta_4 = \frac{1}{\Gamma(M\mu_r)}$, $\beta_5 = M\mu_r$, $\beta_6 = \beta_4(\beta_5 - 1)!$, $\beta_{54} = k_1\bar{\alpha}_r$, $\beta_{55} = \frac{\beta_6\beta_3^{k_1}}{k_1!}$.

B. Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of $R - D$ link

1) *PDF of $R - D$ link:* The PDF of the $R - D$ link can be obtained as [4],

$$f_{rd}(x) = \delta_1 x^{\delta_2} e^{-\delta_3 x^{\bar{\alpha}_d}}, \quad (4)$$

where $\delta_1 = \frac{\bar{\alpha}_d (M_1\mu_d)^{M_1\mu_d}}{\Gamma(M_1\mu_d)(M_1\bar{x}_d)^{M_1\mu_d\bar{\alpha}_d}}$, $\delta_2 = M_1\mu_d\bar{\alpha}_d - 1$, $\delta_3 = \frac{M_1\mu_d}{M_1\bar{\alpha}_d\bar{x}_d^{\bar{\alpha}_d}}$, α_d is the fading parameter of the $R - D$ link, M_1 is the number of receiving antennas at destination, μ_d is the inverse of normalized variance $R - D$ Link, x instantaneous SNR of $R - D$ link, $\bar{x}_d$ average SNR of $R - D$ link, and $\bar{\alpha}_d = \frac{\alpha_d}{2}$.

2) *CDF of $R-D$ link:* The CDF of the the $R - D$ link can be obtained as [4],

$$F_{rd}(x) = \int_0^x f_d(x) dx, \quad (5)$$

Now substituting (4) in (5) and performing the integration using [31, Eq. 3.381.8], we get

$$F_{rd}(x) = \delta_6 - \sum_{k_2=0}^{\delta_5-1} \beta_{13} x^{\beta_{14}} e^{-\delta_3 x^{\bar{\alpha}_d}}, \quad (6)$$

where $\delta_4 = \frac{1}{\Gamma(M_1\mu_d)}$, $\delta_6 = \delta_4(\delta_5 - 1)!$, $\beta_{13} = \frac{\delta_6\delta_3^{k_2}}{k_2!}$, $\beta_{14} = k_2\bar{\alpha}_d$,

C. Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of $S - D$ link

1) *CDF of $S-D$ link:*

$$F_{sd}(x) = F_{rd}(x)(-F_{sr}(x)) + F_{rd}(x) + F_{sr}(x), \quad (7)$$

2) *PDF of $S-D$ link:*

$$f_{sd}(x) = \frac{dF_{sd}(x)}{dx}, \quad (8)$$

D. Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of $S - E_k$ link

1) *PDF of $S - E_k$ link:* The PDF of the $S - E_k$ link can be obtained as [4],

$$f_{se}(x) = \beta_8 x^{\beta_9} e^{-\beta_{10} x^{-\lambda}}, \quad (9)$$

where $\lambda = \frac{d}{\nu}$, $Cd = \frac{\pi^{d/2}}{\Gamma(\frac{d}{2}+1)}$, $\Omega_e = \frac{\Gamma(p\mu_e)}{\Gamma(p\mu_e + \frac{2}{\alpha_e})}$, $A_{e0} = \frac{\lambda(Cd)Z_e\Omega_e\Gamma(\frac{2\lambda}{\alpha_e} + p\mu_e)}{\Gamma(p\mu_e)}$, $A_{e1} = \frac{A_{e0}}{\lambda}$, $\beta_8 = \frac{\lambda A_{e1}^k \eta_e^{\lambda k}}{\Gamma(k)}$, $\beta_9 = k(-\lambda) - 1$, $\beta_{10} = A_{e1}\eta_e^\lambda$, d is the dimension of Euclidean Space, ν is the path loss exponent, Z_e is the density of eavesdroppers, α_e is the fading parameter of $S - E_k$ link, μ_e is the inverse of normalized variance of $S - E_k$ link, p is the number of antennas of each eavesdroppers, k is the number of eavesdroppers, η_e is the SNR per bandwidth

2) *CDF of $S - E_k$ link:* The CDF of the the $S - E_k$ link can be obtained as [4],

$$F_{se}(x) = \sum_{k_3=0}^{k-1} \beta_{20} x^{-\lambda k_3} e^{-\beta_{21} x^{-\lambda}}, \quad (10)$$

where $\beta_{20} = \frac{((k-1)!(A_{e1})^{k_3} \eta_e^{\lambda k_3})}{k_3! \Gamma(k)}$, $\beta_{21} = A_{e1}\eta_e^\lambda$

E. Average Secrecy Capacity (ASC) Equation for Scenario I

ASC can be given as,

$$ASC = \int_0^\infty \frac{F_{se}(x)}{1+x} [1 - F_{sd}(x)] dx, \quad (11)$$

where $F_{se}(x)$ is the CDF of source to eavesdropper and $F_{sd}(x)$ is the CDF of source to destination. Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get [Appendix 1]

$$ASC_1 = -I_1 - I_2 - I_3 + I_4 \quad (12)$$

F. Lower Bound Secrecy Outage Probability (SOP) Equation for Scenario I

SOP can be given as,

$$SOP_1 = \int_0^\infty F_{sd}(x) f_{se}(x) dx, \quad (13)$$

where $F_{sd}(x)$ is the CDF of source to destination and $f_{se}(x)$ is the PDF of source to eavesdropper. Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get [Appendix 1]

$$SOP_1 = I_5 + I_6 + I_7 - I_8 \quad (14)$$

G. Strictly Positive Secrecy Capacity (SPSC) Equation for Scenario I

SPSC can be given as,

$$SPSC_1 = \int_0^\infty f_{rd}(x) F_{se}(x) dx, \quad (15)$$

where $f_{rd}(x)$ is the PDF of relay to destination and $F_{se}(x)$ is the CDF of source to eavesdropper. [Appendix 1]

H. Intercept Probability (IP) Equation for Scenario I

IP can be given as,

$$IP_1 = 1 - SPSC_1 \quad (16)$$

I. Effective Secrecy Throughput (EST) Equation for Scenario I

EST can be given as,

$$EST_1 = R \times (1 - SOP_1) \quad (17)$$

where R is the target rate used in SOP equation

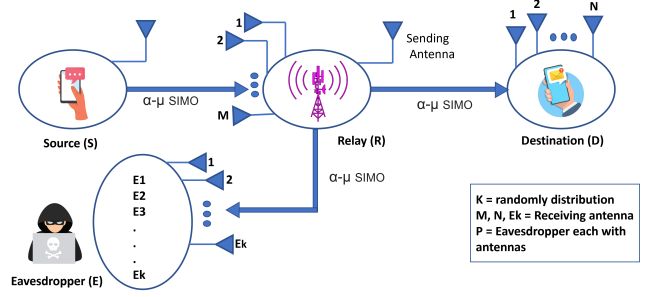


Fig. 4. Eavesdropping in relay

IV. SCENARIO-II

A. Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of $R - E_k$ link

1) *PDF of $R - E_k$ link:* The PDF of the $R - E_k$ link can be obtained as [4],

$$f_{re}(x) = \beta_{51} x^{\beta_{52}} e^{-\beta_{53} x^{-\lambda_{re}}} \quad (18)$$

where $\lambda_{re} = \frac{d_{re}}{\nu_{re}}$, $Cd_{re} = \frac{\pi^{d_{re}/2}}{\Gamma(\frac{d_{re}}{2} + 1)}$, $\Omega_{re} = \frac{\Gamma(p\mu_{re})}{\Gamma(p_{re}\mu_{re} + \frac{2}{\alpha_{re}})}$, $A_{re0} = \frac{\lambda_{re} Cd_{re} z_{re} \Omega_{re}^{\lambda_{re}} \Gamma(\frac{2\lambda_{re}}{\alpha_{re}} + p_{re}\mu_{re})}{\Gamma(p_{re}\mu_{re})}$, $A_{re1} = \frac{A_{re0}}{\lambda_{re}}$, $\beta_{51} = \frac{\lambda_{re} A_{re1} \eta_e^{\lambda_{re} k}}{\Gamma(k_5)}$, $\beta_{52} = k_5(-\lambda_{re}) - 1$, $\beta_{53} = A_{re1} \eta_e^{\lambda_{re}}$, d_{re} is the dimension of Euclidean Space, ν_{re} is the path loss exponent, z_{re} is the density of eavesdroppers, α_{re} is the fading parameter of $R - E_k$ link, μ_{re} is the inverse of normalized variance of $R - E_k$ link, p_{re} is the number of antennas of each eavesdroppers, k_5 is the number of eavesdroppers, $\eta_e f$ is the SNR per bandwidth

2) *CDF of $R - E_k$ link:* The CDF of the $R - E_k$ link can be obtained as [4],

$$F_{re}(x) = \sum_{k_5=0}^{k-1} \beta_{56} x^{\beta_{58}} e^{-\beta_{57} x^{-\lambda_{re}}}, \quad (19)$$

where $\beta_{56} = \frac{(k-1)! (A_{re1})^{k_5} \eta_e^{\lambda_{re} k_5}}{k_5! \Gamma(k)}$, $\beta_{57} = A_{re1} \eta_e^{\lambda_{re}}$, $\beta_{58} = -\lambda_{re} k_5$

B. Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of $S - R$ link

1) *PDF of $S - R$ link:* The PDF of the $S - R$ link is given in equation 1.

2) *CDF of $S - R$ link:* The CDF of the $S - R$ link is given in equation 3.

C. Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of $S - E$ link

1) *CDF of $S-E$ link:*

$$F_{se}(x) = F_{sr}(x) + F_{re}(x) - F_{sr}(x)F_{re}(x), \quad (20)$$

2) *PDF of $S-E$ link:*

$$f_{se}(x) = f_{sr}(x) + f_{re}(x) - F_{sr}(x)f_{re}(x) - f_{sr}(x)F_{re}(x), \quad (21)$$

D. Average Secrecy Capacity (ASC) Equation for Scenario II

ASC can be given as from equation 11,

where $F_{se}(x)$ is the CDF of source to eavesdropper and $F_{sd}(x)$ is the CDF of source to destination. [Appendix 2]

$$ASC_2 = -I_9 - I_{10} - I_{11} - I_{12} + I_{13} - I_{14} - I_{15} + I_{16} - I_{17} - I_{18} - I_{19} + I_{20} \quad (22)$$

E. Lower Bound Secrecy Outage Probability (SOP) Equation for Scenario II

SOP can be given as from equation 31, where $F_{sd}(x)$ is the CDF of source to destination and $f_{se}(x)$ is the PDF of source to eavesdropper. Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get [Appendix 2]

$$SOP_2 = I_{21} + I_{22} - I_{23} + I_{24} + I_{25} + I_{26} - I_{27} + I_{28} + I_{29} + I_{30} - I_{31} + I_{32} - I_{33} + I_{34} + I_{35} - I_{36} \quad (23)$$

F. Strictly Positive Secrecy Capacity (SPSC) Equation for Scenario II

SPSC can be given as,

$$SPSC_2 = 1 - \int_0^\infty F_{rd}(x)f_{re}(x) dx, \quad (24)$$

$$= 1 - (I_{42} - I_{43}) \quad (25)$$

where $F_{rd}(x)$ is the CDF of relay to destination and $f_{re}(x)$ is the PDF of relay to eavesdropper. [Appendix 2]

G. Intercept Probability (IP) Equation for Scenario II

SPSC can be given as,

$$IP_2 = 1 - SPSC_2 \quad (26)$$

H. Effective Secrecy Throughout (EST) Equation for Scenario II

EST can be given as,

$$EST_2 = R \times (1 - SOP_2) \quad (27)$$

where R is the target rate used in SOP equation

V. SCENARIO-III (ATTACK IN BOTH SOURCE AND RELAY)

A. Lower Bound Secrecy Outage Probability (SOP) Equation for Scenario III

When both the source and relay hops are simultaneously vulnerable to wiretapping, an outage occurs if either of the individual links fails to sustain the target secrecy rate R_s . Consequently, the lower bound on the secrecy outage probability can be expressed as:

$$SOP_3 = 1 - (1 - S_1)(1 - S_2) = S_1 + S_2 - S_1S_2, \quad (28)$$

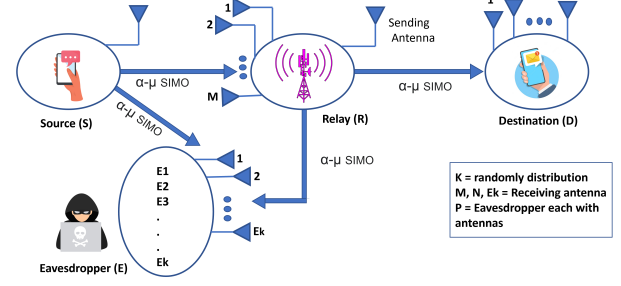


Fig. 5. Simultaneous eavesdropping attacks on both source and relay hops.

where S_1 and S_2 denote the individual outage probabilities of the first and second hops, respectively, defined as:

$$S_1 = \int_0^\infty F_{sr}(a_3x)f_{se}(x) dx, \quad (29)$$

$$S_2 = \int_0^\infty F_{rd}(a_3x)f_{re}(x) dx, \quad (30)$$

where $F_{sr}(\cdot)$ is the CDF of the $S - R$ link given in (6), $f_{se}(\cdot)$ is the PDF of the $S - E_k$ wiretap link given in (9), $F_{rd}(\cdot)$ is the CDF of the $R - D$ link given in (6), and $f_{re}(\cdot)$ is the PDF of the $R - E_k$ link given in (18). Here, $a_3 = 2^{2R_s}$ denotes the threshold factor corresponding to the predetermined secrecy rate R_s , and the transformation parameters are defined as $\theta_1 = \theta_2 = 2^{2R_s}$.

Evaluating these integrals yields:

$$S_1 = I_{37} - I_{38}, \quad (31)$$

$$S_2 = I_{39} - I_{40}, \quad (32)$$

whose detailed closed-form solutions are developed in Appendix X.

B. Strictly Positive Secrecy Capacity (SPSC) Equation for Scenario III

The strictly positive secrecy capacity for Scenario-III requires non-zero positive capacity simultaneously on both hops:

$$SPSC_3 = (1 - \mu_1)(1 - \mu_2), \quad (33)$$

where:

$$\mu_1 = \int_0^\infty F_{sr}(x)f_{se}(x) dx = I_{44} - I_{45}, \quad (34)$$

$$\mu_2 = \int_0^\infty F_{rd}(x)f_{re}(x) dx = I_{46} - I_{47}. \quad (35)$$

C. Intercept Probability (IP) Equation for Scenario III

The intercept probability is obtained from the complement of the SPSC metric:

$$IP_3 = 1 - SPSC_3. \quad (36)$$

D. Effective Secrecy Throughput (EST) Equation for Scenario III

The effective secrecy throughput evaluates the average transmitted confidential rate penalty due to outages:

$$\text{EST}_3 = R_s \times (1 - \text{SOP}_3), \quad (37)$$

where R_s denotes the target transmission secrecy rate.

VI. NUMERICAL RESULTS AND DISCUSSION

In this section, the analytical expressions derived for the secrecy performance metrics—namely secrecy outage probability (SOP), strictly positive secrecy capacity (SPSC), intercept probability (IP), and effective secrecy throughput (EST)—are evaluated and verified across Scenarios I, II, and III. Unless specified otherwise, the default parameters configured for the numerical evaluations are: path loss exponent $\nu = 3.5$, dimension of the Euclidean space $d = 2$, eavesdropper spatial density $Z_e = 10^{-3}$ nodes/m², target secrecy transmission rate $R_s = 1.0$ bps/Hz, legitimate receive antenna counts $M = 2$ and $M_1 = 2$, eavesdropper antenna count $p = 1$, and generalized fading parameters $\alpha_r = \alpha_d = \alpha_e = 2$, $\mu_r = \mu_d = \mu_e = 1$.

A. Impact of Legitimate SNR and Multi-Antenna Diversity

Increasing the average legitimate signal-to-noise ratio ($\bar{x}_r, \bar{x}_d$) substantially improves the secrecy performance of the network by reducing the SOP and increasing both the SPSC and EST across all scenarios. Equipping the relay (R) and destination (D) with multiple receiving antennas ($M, M_1 > 1$) yields significant array gain, which suppresses the vulnerability of legitimate transmissions even when eavesdroppers possess multiple antennas.

B. Vulnerability Comparison Across Scenarios

Comparing the three operational models highlights distinct security dynamics:

- **Scenario-I vs. Scenario-II:** When eavesdroppers only intercept either the first hop ($S - R$) or the second hop ($R - D$), the secrecy bottleneck is determined by the weaker link. In asymmetric deployments where the relay possesses higher processing power or more antennas than the destination ($M > M_1$), Scenario-I provides superior physical layer secrecy compared to Scenario-II.
- **Scenario-III (Simultaneous Eavesdropping):** Scenario-III represents the most critical security vulnerability. Because unauthorized receivers concurrently tap both transmission phases, the secrecy outage probability is significantly elevated compared to single-hop interception, demanding higher legitimate transmit power or enhanced antenna diversity to sustain secure throughput.

C. Impact of Eavesdropper Density and Path Loss

As the spatial density of eavesdroppers Z_e increases, the probability of an eavesdropper being located in close proximity to the source or relay rises sharply, driving the intercept

probability (IP) asymptotically toward unity. Furthermore, environments with severe path-loss exponents (ν) reduce the effective interception range of distant eavesdroppers, thereby enhancing the overall secrecy throughput for authorized receivers.

VII. CONCLUSION

This paper investigated the physical layer security performance of a dual-hop SIMO wireless relay network operating over generalized $\alpha - \mu$ fading channels in the presence of stochastically distributed eavesdroppers. Three distinct attack paradigms were formulated: source-hop wiretapping (Scenario-I), relay-hop wiretapping (Scenario-II), and simultaneous dual-hop wiretapping (Scenario-III). Unified analytical expressions were derived for key physical layer security metrics, including the lower bound secrecy outage probability, strictly positive secrecy capacity, intercept probability, and effective secrecy throughput using Meijer's G -functions. The analytical framework provides rigorous tools for dimensioning secure relay networks and demonstrates that simultaneous eavesdropping significantly diminishes secrecy, but deploying multiple legitimate receive antennas effectively mitigates the threat posed by distributed adversaries.

VIII. APPENDIX 1

A. Scenario-I

1) *Average Secrecy Capacity (ASC) Equation for Scenario I:*

$$I_1 = \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} (-1)^{k_7} \beta_{23} \int_0^{\infty} x^{\beta_{22}} e^{-\beta_{21}x^{-\lambda}} dx \quad (38)$$

$$= \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} (-1)^{k_7} \beta_{23} \frac{\Gamma(\nu_4)}{n_4(\beta_{21})^{\nu_4}}, \quad (39)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$I_2 = \sum_{k_1=0}^{\beta_5-1} \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} (-1)^{k_7} \beta_{34} \int_0^{\infty} x^{\beta_{25}} e^{-\beta_{33}x^{\bar{\alpha}_r} - \beta_{21}x^{-\lambda}} dx, \quad (40)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} (-1)^{k_7} \beta_{34} \int_0^{\infty} y^{\beta_{32}-1} G_{0,1}^{1,0} \left[\beta_{33}y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta_{21}y^{-\lambda/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (41)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} \gamma_{10} G_{-\lambda, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda} \left[\gamma_{11} \left| \begin{matrix} \Delta(-\lambda, 1 - \beta_{32}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (42)$$

$$I_3 = \sum_{k_2=0}^{\delta_5-1} \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} (-1)^{k_7} \beta_{35} \int_0^{\infty} x^{\beta_{27}} e^{-\delta_3 x^{\bar{\alpha}_d} - \beta_{21}x^{-\lambda}} dx, \quad (43)$$

$$= \sum_{k_2=0}^{\delta_5-1} \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} (-1)^{k_7} \beta_{35} \int_0^{\infty} y^{\beta_{33}-1} G_{0,1}^{1,0} \left[\delta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\ G_{0,1}^{1,0} \left[\beta_{21} y^{-\lambda/\bar{\alpha}_d} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (44)$$

$$= \sum_{k_2=0}^{\delta_5-1} \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} \gamma_{12} G_{-\lambda, \bar{\alpha}_d}^{\bar{\alpha}_d, -\lambda} \left[\gamma_{13} \left| \begin{matrix} \Delta(-\lambda, 1 - \beta_{33}) \\ \Delta(\bar{\alpha}_d) \end{matrix} \right. \right], \quad (45)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$I_4 = \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} \gamma_{14} \int_0^{\infty} x^{\beta_{31}} e^{-\beta_{33}x^{\bar{\alpha}_r} - \delta_3 x^{\bar{\alpha}_d} - \beta_{21}x^{-\lambda}} dx, \quad (46)$$

let, $\bar{\alpha}_r = \bar{\alpha}_d = -\lambda$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} \gamma_{15} \int_0^{\infty} y^{\gamma_{16}} e^{-\gamma_{17}y} dy, \quad (47)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \sum_{k_3=0}^{k-1} \sum_{k_7=0}^{\infty} \gamma_{18}^{-\left(\frac{\beta_{31}+1-2\bar{\alpha}_r}{\bar{\alpha}_r}\right)}, \quad (48)$$

$$ASC = -I_1 - I_2 - I_3 + I_4 \quad (49)$$

where $\beta_{10} = \beta_6 + \delta_6 - \beta_6\delta_6$, $\beta_{11} = \frac{\beta_6\beta_3^{k_1}}{k_1!}$, $\beta_{12} = k_1\bar{\alpha}_r$, $\beta_{13} = \frac{\delta_6\delta_3^{k_2}}{k_2!}$, $\beta_{14} = k_2\bar{\alpha}_d$, $\beta_{15} = \beta_{11}\delta_6$, $\beta_{16} = \beta_{13}\beta_6$, $\beta_{17} = \beta_{11}\beta_{13}$, $\beta_{18} = \beta_{12} + \beta_{14}$, $\beta_{19} = 1 - \beta_{10}$, $\beta_{20} = \frac{(k-1)!(A_{e1})^{k_3}\eta_e^{\lambda k_3}}{k_3\Gamma(k)}$, $\beta_{21} = A_{e1}\eta_e^{\lambda}$, $\beta_{22} = k_7 - \lambda k_3$, $\beta_{23} = \beta_{19}\beta_{20}$, $\beta_{24} = \beta_{11}\beta_{20}$, $\beta_{25} = \beta_{12}\beta_{22}$, $\beta_{26} = \beta_{13}\beta_{20}$, $\beta_{27} = \beta_{14} + \beta_{22}$, $\beta_{28} = \beta_{15}\beta_{20}$, $\beta_{29} = \beta_{16}\beta_{20}$, $\beta_{30} = \beta_{17}\beta_{20}$, $\beta_{31} = \beta_{18} + \beta_{20}$, $\beta_{32} = \frac{\beta_{25}+1}{\bar{\alpha}_r}$, $\beta_{33} = \frac{\beta_{27}+1}{\bar{\alpha}_d}$, $\beta_{34} = \beta_{24} + \beta_{28}$, $\beta_{35} = \beta_{26} + \beta_{29}$, $m_4 = \beta_{22}$, $n_4 = -\lambda$, $\nu_4 = \frac{m_4+1}{n_4}$, $\gamma_{10} = (-1)^{k_7} \beta_{34} \frac{\bar{\alpha}_r^{0.5}(-\lambda)^{\beta_{32}-0.5}(\beta_3)^{-\beta_{32}}}{(2\pi)^{\frac{\bar{\alpha}_r-\lambda-2}{2}}}$, $\gamma_{11} = \frac{\beta_{21}^{\bar{\alpha}_r}(\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\beta_3^{-\lambda}(-\lambda)^{\lambda}}$, $\gamma_{12} = (-1)^{k_7} \beta_{35} \frac{\bar{\alpha}_d^{0.5}(-\lambda)^{\beta_{33}-0.5}(\delta_3)^{-\beta_{33}}}{(2\pi)^{\frac{\bar{\alpha}_d-\lambda-2}{2}}}$, $\gamma_{13} = \frac{\beta_{21}^{\bar{\alpha}_d}(\bar{\alpha}_d)^{-\bar{\alpha}_d}}{\delta_3^{-\lambda}(-\lambda)^{\lambda}}$, $\gamma_{14} = (-1)^{k_7} \beta_{30}$, $\gamma_{15} = (-1)^{k_7} \frac{\beta_{30}}{\bar{\alpha}_r}$, $\gamma_{16} = (\frac{\beta_{31}+1-\bar{\alpha}_r}{\bar{\alpha}_r})$, $\gamma_{17} = (\beta_3 + \delta_3 + \beta_{21})$, $\gamma_{18} = \gamma_{15}(\gamma_{16}!)\gamma_{17}$.

2) Lower Bound Secrecy Outage Probability (SOP) Equation for Scenario I:

$$I_5 = \alpha_3 \int_0^{\infty} x^{\beta_9} e^{-\beta_{10}x^{-\lambda}} dx \quad (50)$$

$$= \alpha_3 \frac{\Gamma(\nu_5)}{n_5(\beta_{10})^{\nu_5}}, \quad (51)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$I_6 = \sum_{k_1=0}^{\beta_5-1} \alpha_4 \int_0^{\infty} x^{\alpha_5} e^{-\beta_{43}x^{\bar{\alpha}_r} - \beta_{10}x^{-\lambda}} dx, \quad (52)$$

$$= \sum_{k_1=0}^{\beta_5-1} \alpha_4 \int_0^{\infty} y^{\alpha_{10}-1} G_{0,1}^{1,0} \left[\beta_{43} y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\ G_{0,1}^{1,0} \left[\beta_{10} y^{-\lambda/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (53)$$

$$= \sum_{k_1=0}^{\beta_5-1} \gamma_{20} G_{-\lambda, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda} \left[\gamma_{21} \left| \begin{matrix} \Delta(-\lambda, 1 - \alpha_{10}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (54)$$

$$I_7 = \sum_{k_2=0}^{\delta_5-1} \alpha_6 \int_0^{\infty} x^{\alpha_7} e^{-\beta_{45}x^{\bar{\alpha}_d} - \beta_{10}x^{-\lambda}} dx, \quad (55)$$

$$= \sum_{k_2=0}^{\delta_5-1} \alpha_6 \int_0^{\infty} y^{\alpha_{11}-1} G_{0,1}^{1,0} \left[\beta_{45} y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\ G_{0,1}^{1,0} \left[\beta_{10} y^{-\lambda/\bar{\alpha}_d} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (56)$$

$$= \sum_{k_2=0}^{\delta_5-1} \gamma_{22} G_{-\lambda, \bar{\alpha}_d}^{\bar{\alpha}_d, -\lambda} \left[\gamma_{23} \left| \begin{matrix} \Delta(-\lambda, 1 - \alpha_{11}) \\ \Delta(\bar{\alpha}_d) \end{matrix} \right. \right], \quad (57)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$I_8 = \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \alpha_8 \int_0^{\infty} x^{\alpha_9} e^{-\beta_{43}x^{\bar{\alpha}_r} - \beta_{45}x^{\bar{\alpha}_d} - \beta_{10}x^{-\lambda}} dx, \quad (58)$$

let, $\bar{\alpha}_r = \bar{\alpha}_d = -\lambda$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \gamma_{24} \int_0^{\infty} y^{\gamma_{25}} e^{-\gamma_{26}y} dy, \quad (59)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \gamma_{27}^{-\left(\frac{\alpha_9+1-2\bar{\alpha}_r}{\bar{\alpha}_r}\right)}, \quad (60)$$

$$SOP_1 = I_5 + I_6 + I_7 - I_8 \quad (61)$$

where $\beta_{14} = k_2\bar{\alpha}_d$, $\beta_{18} = \beta_{47} + \beta_{14}$, $\beta_{41} = \beta_6 + \delta_6 - \beta_6\delta_6$, $\beta_{42} = \frac{\beta_6\beta_3^{k_1}}{k_1!}(\theta_1)^{k_1\bar{\alpha}_r}$, $\beta_{43} = \beta_3(\theta_1)^{\bar{\alpha}_r}$, $\beta_{44} = \frac{\delta_6\delta_3^{k_2}}{k_2!}(\theta_1)^{k_2\bar{\alpha}_d}$, $\beta_{45} = \delta_3(\theta_1)^{\bar{\alpha}_d}$, $\beta_{46} = \beta_{42}\delta_6$, $\beta_{47} = k_1\bar{\alpha}_r$, $\beta_{48} = \beta_{44}\beta_6$, $\beta_{49} = \beta_{42}\beta_{44}$, $\alpha_3 = \beta_{41}\beta_8$, $\alpha_4 = (\beta_{46} - \beta_{42})\beta_8$, $\alpha_5 = \beta_{47} + \beta_9$, $\alpha_6 = (\beta_{48} - \beta_{44})\beta_8$, $\alpha_7 = \beta_{14} + \beta_9$, $\alpha_8 = \beta_{49}\beta_8$, $\alpha_9 = \beta_{18} + \beta_9$, $\alpha_{10} = \frac{\alpha_5+1}{\bar{\alpha}_r}$, $\alpha_{11} = \frac{\alpha_7+1}{\bar{\alpha}_d}$, $m_5 = \beta_9$, $n_5 = -\lambda$, $\nu_5 = \frac{m_5+1}{n_5}$, $\gamma_{20} = \alpha_4 \frac{\bar{\alpha}_r^{0.5}(-\lambda)^{\alpha_{10}-0.5}(\beta_{43})^{-\alpha_{10}}}{(2\pi)^{\frac{\bar{\alpha}_r-\lambda-2}{2}}}$, $\gamma_{21} = \frac{\beta_{10}^{\bar{\alpha}_r}(\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\beta_{43}^{-\lambda}(-\lambda)^{\lambda}}$, $\gamma_{22} = \alpha_6 \frac{\bar{\alpha}_d^{0.5}(-\lambda)^{\alpha_{11}-0.5}(\beta_{45})^{-\alpha_{11}}}{(2\pi)^{\frac{\bar{\alpha}_d-\lambda-2}{2}}}$, $\gamma_{23} = \frac{\beta_{10}^{\bar{\alpha}_d}(\bar{\alpha}_d)^{-\bar{\alpha}_d}}{\beta_{45}^{-\lambda}(-\lambda)^{\lambda}}$, $\gamma_{24} = \frac{\alpha_8}{\bar{\alpha}_r}$, $\gamma_{25} = (\frac{\alpha_9+1-\bar{\alpha}_r}{\bar{\alpha}_r})$, $\gamma_{26} = (\beta_{43} + \beta_{45} + \beta_{10})$, $\gamma_{27} = \gamma_{24}(\gamma_{25}!)\gamma_{26}$.

3) *Strictly Positive Secrecy Capacity (SPSC) Equation for Scenario I:*

$$I_{41} = \sum_{k_3=0}^{k-1} \gamma_1 \int_0^\infty x^{\gamma_2} e^{-\delta_3 x^{\bar{\alpha}_d} - \beta_{21} x^{-\lambda}} dx, \quad (62)$$

$$= \sum_{k_3=0}^{k-1} \gamma_1 \int_0^\infty y^{\gamma_1-1} G_{0,1}^{1,0} \left[\delta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\ G_{0,1}^{1,0} \left[\beta_{21} y^{-\lambda/\bar{\alpha}_d} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (63)$$

$$= \sum_{k_3=0}^{k-1} \gamma_1 \frac{\bar{\alpha}_d^{0.5} (-\lambda)^{\gamma_3-0.5} (\delta_3)^{-\gamma_3}}{(2\pi)^{\frac{\bar{\alpha}_d-\lambda-2}{2}}} \times \\ G_{-\lambda, \bar{\alpha}_d}^{\bar{\alpha}_d, -\lambda} \left[\frac{\beta_{21}^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}{\delta_3^{-\lambda} (-\lambda)^\lambda} \left| \begin{matrix} \Delta(-\lambda, 1-\gamma_3) \\ \Delta(\bar{\alpha}_d) \end{matrix} \right. \right], \quad (64)$$

where $\gamma_1 = \delta_1 \beta_{20}$, $\gamma_2 = \delta_2 - k_3 \lambda$, $\gamma_3 = \frac{\gamma_2+1}{\bar{\alpha}_d}$,

IX. APPENDIX 2

A. Scenario-II

1) *Average Secrecy Capacity (ASC) Equation for Scenario II:*

$$I_9 = \quad (65)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$I_{10} = \sum_{k_1=0}^{\beta_5-1} \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{67} \int_0^\infty x^{\beta_{83}} e^{-\beta_3 x^{\bar{\alpha}_r}} dx \quad (66)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{67} \frac{\Gamma(\nu_1)}{n_1 (\beta_3)^{\nu_1}}, \quad (67)$$

$$I_{11} = \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{68} \int_0^\infty x^{\beta_{84}} e^{-\beta_{57} x^{-\lambda_{re}}} dx \quad (68)$$

$$= \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{68} \frac{\Gamma(\nu_2)}{n_2 (\beta_{87})^{\nu_2}}, \quad (69)$$

$$I_{13} = \left(\sum_{k_1=0}^{\beta_5-1} \right) \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{70} \int_0^\infty x^{\beta_{86}} e^{-2\beta_3 x^{\bar{\alpha}_r}} dx \quad (70)$$

$$= \left(\sum_{k_1=0}^{\beta_5-1} \right) \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{70} \frac{\Gamma(\nu_3)}{n_3 (2\beta_3)^{\nu_3}}, \quad (71)$$

$$I_{15} = \sum_{k_2=0}^{\delta_5-1} \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{73} \int_0^\infty x^{\beta_{87}} e^{-\delta_3 x^{\bar{\alpha}_d}} dx \quad (72)$$

$$= \sum_{k_2=0}^{\delta_5-1} \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{73} \frac{\Gamma(\nu_6)}{n_6 (\delta_3)^{\nu_6}}, \quad (73)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$I_{12} = \sum_{k_1=0}^{\beta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{28} \int_0^\infty x^{\beta_{85}} e^{-\beta_3 x^{\bar{\alpha}_r} - \beta_{57} x^{-\lambda_{re}}} dx, \quad (74)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{28} \int_0^\infty y^{\beta_{89}-1} G_{0,1}^{1,0} \left[\beta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\ G_{0,1}^{1,0} \left[\beta_{57} y^{-\lambda_{re}/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (75)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{29} G_{-\lambda_{re}, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda_{re}} \left[\gamma_{30} \left| \begin{matrix} \Delta(-\lambda_{re}, 1-\beta_{89}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (76)$$

$$I_{14} = \left(\sum_{k_1=0}^{\beta_5-1} \right) \sum_{k_5=0}^2 \sum_{k_6=0}^{k-1} \gamma_{31} \int_0^\infty x^{\beta_{72}} e^{-2\beta_3 x^{\bar{\alpha}_r} - \beta_{57} x^{-\lambda_{re}}} dx, \quad (77)$$

$$= \left(\sum_{k_1=0}^{\beta_5-1} \right) \sum_{k_5=0}^2 \sum_{k_6=0}^{k-1} \gamma_{31} \int_0^\infty y^{\beta_{90}-1} G_{0,1}^{1,0} \left[2\beta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\ G_{0,1}^{1,0} \left[\beta_{57} y^{-\lambda_{re}/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (78)$$

$$= \left(\sum_{k_1=0}^{\beta_5-1} \right) \sum_{k_5=0}^2 \sum_{k_6=0}^{k-1} \gamma_{32} G_{-\lambda_{re}, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda_{re}} \left[\gamma_{33} \left| \begin{matrix} \Delta(-\lambda_{re}, 1-\beta_{90}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (79)$$

$$I_{16} = \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \sum_{k_6=0}^{\infty} \gamma_{34} \int_0^\infty x^{\beta_{88}} e^{-\beta_3 x^{\bar{\alpha}_r} - \delta_3 x^{\bar{\alpha}_d}} dx, \quad (80)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \sum_{k_6=0}^{\infty} \gamma_{34} \int_0^\infty y^{\beta_{91}-1} G_{0,1}^{1,0} \left[\beta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\ G_{0,1}^{1,0} \left[\delta_3 y^{\gamma_{35}} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (81)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \sum_{k_6=0}^{\infty} \gamma_{36} G_{\bar{\alpha}_d, \bar{\alpha}_r}^{\bar{\alpha}_r, \bar{\alpha}_d} \left[\gamma_{37} \left| \begin{matrix} \Delta(\bar{\alpha}_d, 1-\beta_{91}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (82)$$

$$I_{17} = \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{38} \int_0^\infty x^{\beta_{76}} e^{-\delta_3 x^{\bar{\alpha}_d} - \beta_{57} x^{-\lambda_{re}}} dx, \quad (83)$$

$$= \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{38} \int_0^{\infty} y^{\beta_{92}-1} G_{0,1}^{1,0} \left[\delta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta_{57} y^{\gamma_{39}} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (84)$$

$$= \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{40} G_{-\lambda_{re}, \bar{\alpha}_d}^{\bar{\alpha}_d, -\lambda_{re}} \left[\gamma_{41} \left| \begin{matrix} \Delta(-\lambda_{re}, 1 - \beta_{92}) \\ \Delta(\bar{\alpha}_d) \end{matrix} \right. \right], \quad (85)$$

$$I_{19} = \left(\sum_{k_1=0}^{\beta_5-1} \right)^2 \sum_{k_2=0}^{\delta_5-1} \sum_{k_6=0}^{\infty} \gamma_{42} \int_0^{\infty} x^{\beta_{80}} e^{-2\beta_3 x^{\bar{\alpha}_r} - \delta_3 x^{\bar{\alpha}_d}} dx, \quad (86)$$

$$= \left(\sum_{k_1=0}^{\beta_5-1} \right)^2 \sum_{k_2=0}^{\delta_5-1} \sum_{k_6=0}^{\infty} \gamma_{42} \int_0^{\infty} y^{\beta_{93}-1} G_{0,1}^{1,0} \left[2\beta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\delta_3 y^{\bar{\alpha}_d/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (87)$$

$$= \left(\sum_{k_1=0}^{\beta_5-1} \right)^2 \sum_{k_2=0}^{\delta_5-1} \sum_{k_6=0}^{\infty} \gamma_{43} G_{\bar{\alpha}_d, \bar{\alpha}_r}^{\bar{\alpha}_d, \bar{\alpha}_r} \left[\gamma_{44} \left| \begin{matrix} \Delta(\bar{\alpha}_d, 1 - \beta_{93}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (88)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$I_{18} = \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{77} \times \int_0^{\infty} x^{\beta_{78}} e^{-\beta_3 x^{\bar{\alpha}_r} - \delta_3 x^{\bar{\alpha}_d} - \beta_{57} x^{-\lambda_{re}}} dx, \quad (89)$$

let, $\bar{\alpha}_r = \bar{\alpha}_d = -\lambda_{re}$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{45} \int_0^{\infty} y^{\gamma_{46}} e^{-\gamma_{47} y} dy, \quad (90)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{48}^{-\left(\frac{\beta_{78}+1-2\bar{\alpha}_r}{\bar{\alpha}_r}\right)}, \quad (91)$$

$$I_{20} = \left(\sum_{k_1=0}^{\beta_5-1} \right)^2 \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} (-1)^{k_6} \beta_{81} \times \int_0^{\infty} x^{\beta_{82}} e^{-2\beta_3 x^{\bar{\alpha}_r} - \delta_3 x^{\bar{\alpha}_d} - \beta_{57} x^{-\lambda_{re}}} dx, \quad (92)$$

let, $\bar{\alpha}_r = \bar{\alpha}_d = -\lambda_{re}$

$$= \left(\sum_{k_1=0}^{\beta_5-1} \right)^2 \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{50} \int_0^{\infty} y^{\gamma_{51}} e^{-\gamma_{52} y} dy, \quad (93)$$

$$= \left(\sum_{k_1=0}^{\beta_5-1} \right)^2 \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \sum_{k_6=0}^{\infty} \gamma_{53}^{-\left(\frac{\beta_{82}+1-2\bar{\alpha}_r}{\bar{\alpha}_r}\right)}, \quad (94)$$

$$ASC_2 = -I_9 - I_{10} - I_{11} - I_{12} + I_{13} - I_{14} - I_{15} + I_{16} - I_{17} - I_{18} - I_{19} + I_{20} \quad (95)$$

where $\beta_{60} = \beta_6 \beta_{60}$, $\beta_{61} = \beta_{55} \beta_{56}$, $\beta_{62} = \beta_{54} + \beta_{58}$, $\beta_{63} = \beta_{11} - \beta_{15}$, $\beta_{64} = \beta_{13} - \beta_{16}$, $\beta_{65} = \beta_{56} \beta_{60}$, $\beta_{66} = \beta_6 \beta_{19}$, $\beta_{67} = \beta_{63} \beta_6 - \beta_{55} \beta_{19}$, $\beta_{68} = \beta_{65} \beta_{19}$, $\beta_{69} = \beta_{61} \beta_{19} - \beta_{65} \beta_{63}$, $\beta_{70} = \beta_{63} \beta_{55}$, $\beta_{71} = \beta_{61} \beta_{62}$, $\beta_{72} = \beta_{62} + \beta_{12} + k_6$, $\beta_{73} = \beta_{64} \beta_6$, $\beta_{74} = \beta_{17} \beta_6 + \beta_{64} \beta_{55}$, $\beta_{75} = \beta_{64} \beta_{65}$, $\beta_{76} = \beta_{14} + \beta_{58} + k_6$, $\beta_{77} = \beta_{61} \beta_{64} - \beta_{17} \beta_{65}$, $\beta_{78} = \beta_{62} + \beta_{14} + k_6$, $\beta_{79} = \beta_{55} \beta_{17}$, $\beta_{80} = \beta_{18} + \beta_{54} + k_6$, $\beta_{81} = \beta_{61} \beta_{17}$, $\beta_{82} = \beta_{62} + \beta_{18} + k_6$, $\beta_{83} = \beta_{12} + k_6$, $\beta_{84} = \beta_{58} + k_6$, $\beta_{85} = \beta_{62} + k_6$, $\beta_{86} = 2\beta_{12} + k_6$, $\beta_{87} = \beta_{14} + k_6$, $\beta_{88} = \beta_{18} + k_6$, $\beta_{89} = \frac{\beta_{85}+1}{\bar{\alpha}_r}$, $\beta_{90} = \frac{\beta_{72}+1}{\bar{\alpha}_r}$, $\beta_{91} = \frac{\beta_{88}+1}{\bar{\alpha}_r}$, $\beta_{92} = \frac{\beta_{76}+1}{\bar{\alpha}_d}$, $m_1 = \beta_{83}$, $n_1 = \bar{\alpha}_r$, $\nu_1 = \frac{m_1+1}{n_1}$, $m_2 = \beta_{84}$, $n_2 = -\lambda_{re}$, $\nu_2 = \frac{m_2+1}{n_2}$, $m_3 = \beta_{86}$, $n_3 = \bar{\alpha}_r$, $\nu_3 = \frac{m_3+1}{n_3}$, $m_6 = \beta_{87}$, $n_6 = \bar{\alpha}_d$, $\nu_6 = \frac{m_6+1}{n_6}$, $\gamma_{28} = (-1)^{k_6} \beta_{69}$, $\gamma_{29} = \gamma_{28} \frac{\bar{\alpha}_r^{0.5} (-\lambda_{re})^{\beta_{89}-0.5} (\beta_3)^{-\beta_{89}}}{(2\pi)^{\frac{\bar{\alpha}_r - \lambda_{re} - 2}{2}}}$, $\gamma_{30} = \frac{\beta_{57}^{\bar{\alpha}_r} (\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\beta_3^{-\lambda_{re}} (-\lambda_{re})^{\lambda_{re}}}$, $\gamma_{31} = (-1)^{k_6} \beta_{71}$, $\gamma_{32} = \gamma_{31} \frac{\bar{\alpha}_r^{0.5} (-\lambda_{re})^{\beta_{90}-0.5} (2\beta_3)^{-\beta_{90}}}{(2\pi)^{\frac{\bar{\alpha}_r - \lambda_{re} - 2}{2}}}$, $\gamma_{33} = \frac{\beta_{57}^{\bar{\alpha}_r} (\bar{\alpha}_r)^{-\bar{\alpha}_r}}{(2\beta_3)^{-\lambda_{re}} (-\lambda_{re})^{\lambda_{re}}}$, $\gamma_{34} = (-1)^{k_6} \beta_{74}$, $\gamma_{35} = \bar{\alpha}_d / \bar{\alpha}_r$, $\gamma_{36} = \gamma_{34} \frac{\bar{\alpha}_r^{0.5} (\bar{\alpha}_d)^{\beta_{91}-0.5} (\beta_3)^{-\beta_{91}}}{(2\pi)^{\frac{\bar{\alpha}_r + \bar{\alpha}_d - 2}{2}}}$, $\gamma_{37} = \frac{\delta_3^{\bar{\alpha}_r} (\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\beta_3^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}$, $\gamma_{38} = (-1)^{k_6} \beta_{75}$, $\gamma_{39} = -\lambda_{re} / \bar{\alpha}_d$, $\gamma_{40} = \gamma_{38} \frac{\bar{\alpha}_d^{0.5} (-\lambda_{re})^{\beta_{92}-0.5} (\delta_3)^{-\delta_{92}}}{(2\pi)^{\frac{\bar{\alpha}_d - \lambda_{re} - 2}{2}}}$, $\gamma_{41} = \frac{\beta_{57}^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}{\delta_3^{-\lambda_{re}} (-\lambda_{re})^{\lambda_{re}}}$, $\gamma_{42} = (-1)^{k_6} \beta_{79}$, $\gamma_{43} = \gamma_{42} \frac{\bar{\alpha}_r^{0.5} (\bar{\alpha}_d)^{\beta_{93}-0.5} (2\beta_3)^{-\beta_{93}}}{(2\pi)^{\frac{\bar{\alpha}_r + \bar{\alpha}_d - 2}{2}}}$, $\gamma_{44} = \frac{\delta_3^{\bar{\alpha}_r} (\bar{\alpha}_r)^{-\bar{\alpha}_r}}{(2\beta_3)^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}$, $\gamma_{45} = (-1)^{k_6} \frac{\beta_{77}}{\bar{\alpha}_r}$, $\gamma_{46} = \left(\frac{\beta_{78}+1-\bar{\alpha}_r}{\bar{\alpha}_r}\right)$, $\gamma_{47} = (\beta_3 + \delta_3 + \beta_{57})$, $\gamma_{48} = \gamma_{45} (\gamma_{46}!)$, $\gamma_{49} = (-1)^{k_6} \frac{\beta_{81}}{\bar{\alpha}_r}$, $\gamma_{51} = \left(\frac{\beta_{82}+1-\bar{\alpha}_r}{\bar{\alpha}_r}\right)$, $\gamma_{52} = (2\beta_3 + \delta_3 + \beta_{57})$, $\gamma_{53} = \gamma_{50} (\gamma_{51}!) \gamma_{52}$.

2) Lower Bound Secrecy Outage Probability (SOP) Equation for Scenario II:

$$I_{21} = \alpha_{25} \int_0^{\infty} x^{\beta_2} e^{-\beta_3 x^{\bar{\alpha}_r}} dx \quad (96)$$

$$= \alpha_{25} \frac{\Gamma(\nu_7)}{n_7 (\beta_3)^{\nu_7}}, \quad (97)$$

$$I_{22} = \alpha_{64} \int_0^{\infty} x^{\beta_{52}} e^{-\beta_{53} x^{-\lambda_{re}}} dx \quad (98)$$

$$= \alpha_{64} \frac{\Gamma(\nu_8)}{n_8 (\beta_{53})^{\nu_8}}, \quad (99)$$

$$I_{25} = \sum_{k_1=0}^{\beta_5-1} \alpha_{30} \int_0^{\infty} x^{\alpha_{31}} e^{-\alpha_{36} x^{\bar{\alpha}_r}} dx \quad (100)$$

$$= \sum_{k_1=0}^{\beta_5-1} \alpha_{30} \frac{\Gamma(\nu_9)}{n_9 (\alpha_{31})^{\nu_9}}, \quad (101)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$I_{23} = \sum_{k_5=0}^{k-1} \alpha_{27} \int_0^\infty x^{\alpha_{20}} e^{-\beta_3 x^{\bar{\alpha}_r} - \beta_{53} x^{-\lambda_{re}}} dx, \quad (102)$$

$$= \sum_{k_5=0}^{k-1} \alpha_{27} \int_0^\infty y^{\alpha_{68}-1} G_{0,1}^{1,0} \left[\beta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta_{53} y^{-\lambda_{re}/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (103)$$

$$= \sum_{k_5=0}^{k-1} \gamma_{54} G_{-\lambda_{re}, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda_{re}} \left[\gamma_{55} \left| \begin{matrix} \Delta(-\lambda_{re}, 1 - \alpha_{68}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (104)$$

$$I_{24} = \sum_{k_1=0}^{\beta_5-1} \alpha_{29} \int_0^\infty x^{\alpha_{24}} e^{-\beta_3 x^{\bar{\alpha}_r} - \beta_{53} x^{-\lambda_{re}}} dx, \quad (105)$$

$$= \sum_{k_1=0}^{\beta_5-1} \alpha_{29} \int_0^\infty y^{\alpha_{69}-1} G_{0,1}^{1,0} \left[\beta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta_{53} y^{-\lambda_{re}/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (106)$$

$$= \sum_{k_1=0}^{\beta_5-1} \gamma_{56} G_{-\lambda_{re}, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda_{re}} \left[\gamma_{57} \left| \begin{matrix} \Delta(-\lambda_{re}, 1 - \alpha_{69}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (107)$$

$$I_{26} = \sum_{k_1=0}^{\beta_5-1} \alpha_{65} \int_0^\infty x^{\alpha_{33}} e^{-\beta'_{43} x^{\bar{\alpha}_r} - \beta_{53} x^{-\lambda_{re}}} dx, \quad (108)$$

$$= \sum_{k_1=0}^{\beta_5-1} \alpha_{65} \int_0^\infty y^{\alpha_{70}-1} G_{0,1}^{1,0} \left[\beta'_{43} y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta_{53} y^{-\lambda_{re}/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (109)$$

$$= \sum_{k_1=0}^{\beta_5-1} \gamma_{58} G_{-\lambda_{re}, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda_{re}} \left[\gamma_{59} \left| \begin{matrix} \Delta(-\lambda_{re}, 1 - \alpha_{70}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (110)$$

$$I_{27} = \sum_{k_1=0}^{\beta_5-1} \sum_{k_5=0}^{k-1} \alpha_{34} \int_0^\infty x^{\alpha_{35}} e^{-\alpha_{36} x^{\bar{\alpha}_r} - \beta_{53} x^{-\lambda_{re}}} dx, \quad (111)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_5=0}^{k-1} \alpha_{34} \int_0^\infty y^{\alpha_{71}-1} G_{0,1}^{1,0} \left[\alpha_{36} y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta_{53} y^{-\lambda_{re}/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (112)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_5=0}^{k-1} \gamma_{60} G_{-\lambda_{re}, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda_{re}} \left[\gamma_{61} \left| \begin{matrix} \Delta(-\lambda_{re}, 1 - \alpha_{71}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (113)$$

$$I_{28} = \left(\sum_{k_1=0}^{\beta_5-1} \right)^2 \alpha_{39} \int_0^\infty x^{\alpha_{40}} e^{-\alpha_{36} x^{\bar{\alpha}_r} - \beta_{53} x^{-\lambda_{re}}} dx, \quad (114)$$

$$= \left(\sum_{k_1=0}^{\beta_5-1} \right)^2 \alpha_{39} \int_0^\infty y^{\alpha_{72}-1} G_{0,1}^{1,0} \left[\alpha_{36} y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta_{53} y^{-\lambda_{re}/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (115)$$

$$= \left(\sum_{k_1=0}^{\beta_5-1} \right)^2 \gamma_{62} G_{-\lambda_{re}, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda_{re}} \left[\gamma_{63} \left| \begin{matrix} \Delta(-\lambda_{re}, 1 - \alpha_{72}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (116)$$

$$I_{29} = \sum_{k_2=0}^{\delta_5-1} \alpha_{41} \int_0^\infty x^{\alpha_{42}} e^{-\beta_3 x^{\bar{\alpha}_r} - \beta'_{45} x^{\bar{\alpha}_d}} dx, \quad (117)$$

$$= \sum_{k_2=0}^{\delta_5-1} \alpha_{41} \int_0^\infty y^{\alpha_{73}-1} G_{0,1}^{1,0} \left[\beta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta'_{45} y^{\bar{\alpha}_d/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (118)$$

$$= \sum_{k_2=0}^{\delta_5-1} \gamma_{64} G_{\bar{\alpha}_d, \bar{\alpha}_r}^{\bar{\alpha}_r, \bar{\alpha}_d} \left[\gamma_{65} \left| \begin{matrix} \Delta(\bar{\alpha}_d, 1 - \alpha_{73}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (119)$$

$$I_{30} = \sum_{k_2=0}^{\delta_5-1} \alpha_{66} \int_0^\infty x^{\alpha_{44}} e^{-\beta'_{45} x^{\bar{\alpha}_d} - \beta_{53} x^{-\lambda_{re}}} dx, \quad (120)$$

$$= \sum_{k_2=0}^{\delta_5-1} \alpha_{66} \int_0^\infty y^{\alpha_{74}-1} G_{0,1}^{1,0} \left[\beta'_{45} y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta_{53} y^{-\lambda_{re}/\bar{\alpha}_d} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (121)$$

$$= \sum_{k_2=0}^{\delta_5-1} \gamma_{66} G_{-\lambda_{re}, \bar{\alpha}_d}^{\bar{\alpha}_d, -\lambda_{re}} \left[\gamma_{67} \left| \begin{matrix} \Delta(-\lambda_{re}, 1 - \alpha_{74}) \\ \Delta(\bar{\alpha}_d) \end{matrix} \right. \right], \quad (122)$$

$$I_{33} = \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \alpha_{51} \int_0^\infty x^{\alpha_{52}} e^{-\alpha_{53} x^{\bar{\alpha}_r} - \beta'_{45} x^{\bar{\alpha}_d}} dx, \quad (123)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \alpha_{51} \int_0^\infty y^{\alpha_{75}-1} G_{0,1}^{1,0} \left[\alpha_{53} y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta'_{45} y^{\bar{\alpha}_d/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (124)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \gamma_{68} G_{\bar{\alpha}_d, \bar{\alpha}_r}^{\bar{\alpha}_r, \bar{\alpha}_d} \left[\gamma_{69} \left| \begin{matrix} \Delta(\bar{\alpha}_d, 1 - \alpha_{75}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (125)$$

Now substituting (1) in (2) and performing the integration using [31, Eq. 3.381.8], we get

$$I_{31} = \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \alpha_{45} \int_0^\infty x^{\alpha_{46}} e^{-\beta_{33}x^{\bar{\alpha}_r} - \beta'_{45}x^{\bar{\alpha}_d} - \beta_{53}x^{-\lambda_{re}}} dx, \quad (126)$$

let, $\bar{\alpha}_r = \bar{\alpha}_d = -\lambda_{re}$

$$= \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \frac{\alpha_{45}}{\bar{\alpha}_r} \int_0^\infty y^{\gamma_{70}} e^{-\gamma_{71}y} dy, \quad (127)$$

$$= \sum_{k_2=0}^{\delta_5-1} \sum_{k_5=0}^{k-1} \gamma_{72}^{-\left(\frac{\alpha_{46}+1-2\bar{\alpha}_r}{\bar{\alpha}_r}\right)}, \quad (128)$$

$$I_{32} = \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \alpha_{49} \int_0^\infty x^{\alpha_{50}} e^{-\beta_{33}x^{\bar{\alpha}_r} - \beta'_{45}x^{\bar{\alpha}_d} - \beta_{53}x^{-\lambda_{re}}} dx, \quad (129)$$

let, $\bar{\alpha}_r = \bar{\alpha}_d = -\lambda_{re}$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \frac{\alpha_{49}}{\bar{\alpha}_r} \int_0^\infty y^{\gamma_{73}} e^{-\gamma_{74}y} dy, \quad (130)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \gamma_{75}^{-\left(\frac{\alpha_{50}+1-2\bar{\alpha}_r}{\bar{\alpha}_r}\right)}, \quad (131)$$

$$I_{34} = \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \alpha_{67} \int_0^\infty x^{\alpha_{55}} e^{-\beta'_{43}x^{\bar{\alpha}_r} - \beta'_{45}x^{\bar{\alpha}_d} - \beta_{53}x^{-\lambda_{re}}} dx, \quad (132)$$

let, $\bar{\alpha}_r = \bar{\alpha}_d = -\lambda_{re}$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \frac{\alpha_{67}}{\bar{\alpha}_r} \int_0^\infty y^{\gamma_{76}} e^{-\gamma_{77}y} dy, \quad (133)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \gamma_{78}^{-\left(\frac{\alpha_{55}+1-2\bar{\alpha}_r}{\bar{\alpha}_r}\right)}, \quad (134)$$

$$I_{35} = \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \alpha_{56} \int_0^\infty x^{\alpha_{57}} e^{-\alpha_{58}x^{\bar{\alpha}_r} - \beta'_{45}x^{\bar{\alpha}_d} - \beta_{53}x^{-\lambda_{re}}} dx, \quad (135)$$

let, $\bar{\alpha}_r = \bar{\alpha}_d = -\lambda_{re}$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \frac{\alpha_{56}}{\bar{\alpha}_r} \int_0^\infty y^{\gamma_{79}} e^{-\alpha_{58}x^{\bar{\alpha}_r} - \beta'_{45}x^{\bar{\alpha}_d} - \beta_{53}x^{-\lambda_{re}}} dy, \quad (136)$$

$$= \sum_{k_1=0}^{\beta_5-1} \sum_{k_2=0}^{\delta_5-1} \gamma_{80}^{-\left(\frac{\alpha_{57}+1-2\bar{\alpha}_r}{\bar{\alpha}_r}\right)}, \quad (137)$$

$$I_{36} = \left(\sum_{k_1=0}^{\beta_5-1}\right)^2 \sum_{k_2=0}^{\delta_5-1} \alpha_{61} \int_0^\infty x^{\alpha_{62}} e^{-\alpha_{63}x^{\bar{\alpha}_r} - \beta'_{45}x^{\bar{\alpha}_d} - \beta_{53}x^{-\lambda_{re}}} dx, \quad (138)$$

let, $\bar{\alpha}_r = \bar{\alpha}_d = -\lambda_{re}$

$$= \left(\sum_{k_1=0}^{\beta_5-1}\right)^2 \sum_{k_2=0}^{\delta_5-1} \frac{\alpha_{61}}{\bar{\alpha}_r} \int_0^\infty y^{\gamma_{81}} e^{-\alpha_{63}x^{\bar{\alpha}_r} - \beta'_{45}x^{\bar{\alpha}_d} - \beta_{53}x^{-\lambda_{re}}} dy, \quad (139)$$

$$= \left(\sum_{k_1=0}^{\beta_5-1}\right)^2 \sum_{k_2=0}^{\delta_5-1} \gamma_{82}^{-\left(\frac{\alpha_{62}+1-2\bar{\alpha}_r}{\bar{\alpha}_r}\right)}, \quad (140)$$

$$SOP_2 = I_{21} + I_{22} - I_{23} + I_{24} + I_{25} + I_{26} - I_{27} + I_{28} \\ + I_{29} + I_{30} - I_{31} + I_{32} - I_{33} + I_{34} + I_{35} - I_{36} \quad (141)$$

where $\beta_{14} = k_2\bar{\alpha}_d$, $\beta_{18} = \beta_{47} + \beta_{14}$, $\beta_{41} = \beta_6 + \delta_6 - \beta_6\delta_6$, $\beta'_{42} = \frac{\beta_6\beta_{3k_1}}{k_1!}(\theta_2)^{k_1\bar{\alpha}_r}$, $\beta'_{43} = \beta_3(\theta_2)^{\bar{\alpha}_r}$, $\beta'_{44} = \frac{\delta_6\delta_{3k_2}}{k_2!}(\theta_2)^{k_2\bar{\alpha}_d}$, $\beta'_{45} = \delta_3(\theta_2)^{\bar{\alpha}_d}$, $\beta'_{46} = \beta'_{42}\delta_6$, $\beta_{47} = k_1\bar{\alpha}_r$, $\beta_{48} = \beta'_{44}\beta_6$, $\beta'_{49} = \beta'_{42}\beta'_{44}$, $\beta_{58} = -\lambda_{re}k_5$, $\alpha_{20} = \beta_2 - \beta_{58}$, $\alpha_{21} = \beta_1\beta_{56}$, $\alpha_{22} = \beta_6\beta_{51}$, $\alpha_{23} = \beta_{51}\beta_{55}$, $\alpha_{24} = k_1\bar{\alpha}_r + \beta_{52}$, $\alpha_{25} = \beta_{41}\beta_1$, $\alpha_{26} = \beta_{41}\beta_{51}$, $\alpha_{27} = \beta_{41}\alpha_{21}$, $\alpha_{28} = \beta_{41}\alpha_{22}$, $\alpha_{26} = \beta_{41}\alpha_{23}$, $\alpha_{30} = (\beta'_{46} - \beta'_{42})\beta_1$, $\alpha_{31} = \beta'_{47} + \beta_2$, $\alpha_{32} = (\beta'_{46} - \beta'_{42})\beta_{51}$, $\alpha_{33} = \beta'_{47} + \beta_{52}$, $\alpha_{34} = (\beta'_{46} - \beta'_{42})\alpha_{21}$, $\alpha_{35} = \beta'_{47} + \alpha_{20}$, $\alpha_{36} = \beta'_{43} + \beta_3$, $\alpha_{41} = (\beta'_{48} - \beta'_{44})\beta_1$, $\alpha_{42} = \beta_{14} + \beta_2$, $\alpha_{43} = (\beta'_{48} - \beta'_{44})\beta_{51}$, $\alpha_{44} = \beta_{14} + \beta_{42}$, $\alpha_{45} = (\beta'_{48} - \beta'_{44})\alpha_{21}$, $\alpha_{46} = \beta_{14} + \alpha_{20}$, $\alpha_{47} = (\beta'_{48} - \beta'_{44})\alpha_{22}$, $\alpha_{48} = \beta_{14} + \beta_{52}$, $\alpha_{49} = (\beta'_{48} - \beta'_{44})\alpha_{23}$, $\alpha_{50} = \beta_{14} + \alpha_{24}$, $\alpha_{51} = \beta_1\beta'_{49}$, $\alpha_{52} = \beta_2 + \beta_{18}$, $\alpha_{53} = \beta_3 + \beta'_{43}$, $\alpha_{54} = \beta_{51}\beta'_{49}$, $\alpha_{55} = \beta_{52} + \beta_{18}$, $\alpha_{56} = \alpha_{21}\beta'_{49}$, $\alpha_{57} = \alpha_{20} + \beta_{18}$, $\alpha_{58} = \beta_3 + \beta'_{43}$, $\alpha_{59} = \alpha_{22}\beta'_{49}$, $\alpha_{60} = \beta_{18} + \beta_{52}$, $\alpha_{61} = \alpha_{23}\beta'_{49}$, $\alpha_{62} = \alpha_{24} + \beta_{18}$, $\alpha_{63} = \beta_3 + \beta'_{43}$, $\alpha_{64} = \alpha_{26} - \alpha_{28}$, $\alpha_{65} = \alpha_{32} - \alpha_{37}$, $\alpha_{66} = \alpha_{43} - \alpha_{47}$, $\alpha_{67} = \alpha_{59} - \alpha_{54}$, $\alpha_{68} = \frac{\alpha_{20}+1}{\bar{\alpha}_r}$, $\alpha_{69} = \frac{\alpha_{24}+1}{\bar{\alpha}_r}$, $\alpha_{70} = \frac{\alpha_{33}+1}{\bar{\alpha}_r}$, $\alpha_{71} = \frac{\alpha_{35}+1}{\bar{\alpha}_r}$, $\alpha_{72} = \frac{\alpha_{44}+1}{\bar{\alpha}_r}$, $\alpha_{73} = \frac{\alpha_{42}+1}{\bar{\alpha}_r}$, $\alpha_{74} = \frac{\alpha_{44}+1}{\bar{\alpha}_d}$, $\alpha_{75} = \frac{\alpha_{52}+1}{\bar{\alpha}_r}$, $m_7 = \beta_2$, $n_7 = \bar{\alpha}_r$, $\nu_7 = \frac{m_7+1}{n_7}$, $m_8 = \beta_{52}$, $n_8 = -\lambda_{re}$, $\nu_8 = \frac{m_8+1}{n_8}$, $m_9 = \alpha_{31}$, $n_9 = \bar{\alpha}_r$, $\nu_9 = \frac{m_9+1}{n_9}$, $\gamma_{54} = \frac{\bar{\alpha}_r^{0.5}(-\lambda_{re})^{\alpha_{68}-0.5}(\beta_3)^{-\alpha_{68}}}{(2\pi)^{\frac{\bar{\alpha}_r-\lambda_{re}-2}{2}}}$, $\gamma_{55} = \frac{\beta_{53}^{\bar{\alpha}_r}(\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\beta_3^{-\lambda_{re}}(-\lambda_{re})^{\lambda_{re}}}$, $\gamma_{56} = \frac{\bar{\alpha}_r^{0.5}(-\lambda_{re})^{\alpha_{69}-0.5}(\beta_3)^{-\alpha_{69}}}{(2\pi)^{\frac{\bar{\alpha}_r-\lambda_{re}-2}{2}}}$, $\gamma_{57} = \frac{\beta_{53}^{\bar{\alpha}_r}(\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\beta_3^{-\lambda_{re}}(-\lambda_{re})^{\lambda_{re}}}$, $\gamma_{58} = \frac{\bar{\alpha}_r^{0.5}(-\lambda_{re})^{\alpha_{70}-0.5}(\beta'_{43})^{-\alpha_{70}}}{(2\pi)^{\frac{\bar{\alpha}_r-\lambda_{re}-2}{2}}}$, $\gamma_{59} = \frac{\beta_{53}^{\bar{\alpha}_r}(\bar{\alpha}_r)^{-\bar{\alpha}_r}}{(\beta'_{43})^{-\lambda_{re}}(-\lambda_{re})^{\lambda_{re}}}$, $\gamma_{60} = \frac{\bar{\alpha}_r^{0.5}(-\lambda_{re})^{\alpha_{71}-0.5}(\alpha_{36})^{-\alpha_{71}}}{(2\pi)^{\frac{\bar{\alpha}_r-\lambda_{re}-2}{2}}}$, $\gamma_{61} = \frac{\beta_{53}^{\bar{\alpha}_r}(\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\alpha_{36}^{-\lambda_{re}}(-\lambda_{re})^{\lambda_{re}}}$, $\gamma_{62} = \frac{\bar{\alpha}_r^{0.5}(-\lambda_{re})^{\alpha_{72}-0.5}(\alpha_{36})^{-\alpha_{72}}}{(2\pi)^{\frac{\bar{\alpha}_r-\lambda_{re}-2}{2}}}$, $\gamma_{63} = \frac{\beta_{53}^{\bar{\alpha}_r}(\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\alpha_{36}^{-\lambda_{re}}(-\lambda_{re})^{\lambda_{re}}}$, $\gamma_{64} = \frac{\bar{\alpha}_r^{0.5}(\bar{\alpha}_d)^{\alpha_{73}-0.5}(\beta_3)^{-\alpha_{73}}}{(2\pi)^{\frac{\bar{\alpha}_r+\bar{\alpha}_d-2}{2}}}$,

$$\begin{aligned}
\gamma_{65} &= \frac{(\beta'_{45})^{\bar{\alpha}_r} (\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\beta_3^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}, \quad \gamma_{66} = \alpha_{66} \frac{\bar{\alpha}_d^{0.5} (-\lambda_{re})^{\alpha_{74}-0.5} (\beta'_{45})^{-\alpha_{74}}}{(2\pi)^{\frac{\bar{\alpha}_d - \lambda_{re} - 2}{2}}}, \\
\gamma_{67} &= \frac{\beta_3^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}{(\beta'_{45})^{-\lambda_{re}} (-\lambda_{re})^{\lambda_{re}}}, \quad \gamma_{68} = \alpha_{51} \frac{\bar{\alpha}_r^{0.5} (\bar{\alpha}_d)^{\alpha_{75}-0.5} (\alpha_{53})^{-\alpha_{75}}}{(2\pi)^{\frac{\bar{\alpha}_r + \bar{\alpha}_d - 2}{2}}}, \\
\gamma_{69} &= \frac{(\beta'_{45})^{\bar{\alpha}_r} (\bar{\alpha}_r)^{-\bar{\alpha}_r}}{\alpha_{53}^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}, \quad \gamma_{70} = \left(\frac{\alpha_{46} + 1 - \bar{\alpha}_r}{\bar{\alpha}_r} \right), \quad \gamma_{71} = (\beta_3 + \beta'_{45} + \beta_{53}), \\
\gamma_{72} &= \frac{\alpha_{45}}{\bar{\alpha}_r} (\gamma_{70}!) \gamma_{71}, \quad \gamma_{73} = \left(\frac{\alpha_{50} + 1 - \bar{\alpha}_r}{\bar{\alpha}_r} \right), \\
\gamma_{74} &= (\beta_3 + \beta'_{45} + \beta_{53}), \quad \gamma_{75} = \frac{\alpha_{49}}{\bar{\alpha}_r} (\gamma_{73}!) \gamma_{74}, \quad \gamma_{76} = \left(\frac{\alpha_{55} + 1 - \bar{\alpha}_r}{\bar{\alpha}_r} \right), \\
\gamma_{77} &= (\beta'_{43} + \beta'_{45} + \beta_{53}), \quad \gamma_{78} = \frac{\alpha_{67}}{\bar{\alpha}_r} (\gamma_{76}!) \gamma_{77}, \\
\gamma_{79} &= \left(\frac{\alpha_{57} + 1 - \bar{\alpha}_r}{\bar{\alpha}_r} \right), \quad \gamma_{80} = \frac{\alpha_{56}}{\bar{\alpha}_r} (\gamma_{79}!) (\alpha_{58} + \beta'_{45} + \beta_{53}), \\
\gamma_{81} &= \left(\frac{\alpha_{62} + 1 - \bar{\alpha}_r}{\bar{\alpha}_r} \right), \quad \gamma_{82} = \frac{\alpha_{61}}{\bar{\alpha}_r} (\gamma_{81}!) (\alpha_{63} + \beta'_{45} + \beta_{53}),
\end{aligned}$$

3) *Strictly Positive Secrecy Capacity (SPSC) Equation for Scenario II:*

$$I_{42} = \gamma_4 \int_0^\infty x^{\beta_{52}} e^{-\beta_{53} x^{-\lambda_{re}}} dx \quad (142)$$

$$= \gamma_4 \frac{\Gamma(\nu_{11})}{n_{11} (\beta_{53})^{\nu_{11}}}, \quad (143)$$

$$I_{43} = \sum_{k_2=0}^{\delta_5-1} \gamma_5 \int_0^\infty x^{\gamma_6} e^{-\delta_3 x^{\bar{\alpha}_d} - \beta_{53} x^{-\lambda_{re}}} dx, \quad (144)$$

$$\begin{aligned}
&= \sum_{k_2=0}^{\delta_5-1} \gamma_5 \int_0^\infty y^{\gamma_7-1} G_{0,1}^{1,0} \left[\delta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\
&\quad G_{0,1}^{1,0} \left[\beta_{53} y^{-\lambda_{re}/\bar{\alpha}_d} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (145)
\end{aligned}$$

$$\begin{aligned}
&= \sum_{k_2=0}^{\delta_5-1} \gamma_5 \frac{\bar{\alpha}_d^{0.5} (-\lambda_{re})^{\gamma_7-0.5} (\delta_3)^{-\gamma_7}}{(2\pi)^{\frac{\bar{\alpha}_d - \lambda_{re} - 2}{2}}} \times \\
&\quad G_{-\lambda_{re}, \bar{\alpha}_d}^{\bar{\alpha}_d, -\lambda_{re}} \left[\frac{\beta_3^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}{\delta_3^{-\lambda_{re}} (-\lambda_{re})^{\lambda_{re}}} \left| \begin{matrix} \Delta(-\lambda_{re}, 1 - \gamma_7) \\ \Delta(\bar{\alpha}_d) \end{matrix} \right. \right], \quad (146)
\end{aligned}$$

where $\gamma_4 = \delta_6 \beta_{51}$, $\gamma_5 = \beta_{13} \beta_{51}$, $\gamma_6 = \beta_{14} + \beta_{52}$, $\gamma_7 = \frac{\gamma_6 + 1}{\bar{\alpha}_d}$, $m_{11} = \beta_{52}$, $n_{11} = -\lambda_{re}$, $\nu_{11} = \frac{m_{11} + 1}{n_{11}}$

X. APPENDIX 3

A. Scenario-III

1) *Lower Bound Secrecy Outage Probability (SOP) Equation for Scenario III:*

$$I_{37} = \alpha_{80} \int_0^\infty x^{\beta_9} e^{-\beta_{10} x^{-\lambda}} dx \quad (147)$$

$$= \alpha_{80} \frac{\Gamma(\nu_{10})}{n_{10} (\beta_{53})^{\nu_{10}}}, \quad (148)$$

$$I_{38} = \sum_{k_1=0}^{\beta_5-1} \alpha_{81} \int_0^\infty x^{\alpha_{82}} e^{-\beta'_3 x^{\bar{\alpha}_r} - \beta_{10} x^{-\lambda}} dx, \quad (149)$$

$$\begin{aligned}
&= \sum_{k_1=0}^{\beta_5-1} \alpha_{81} \int_0^\infty y^{\alpha_{83}-1} G_{0,1}^{1,0} \left[\beta'_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\
&\quad G_{0,1}^{1,0} \left[\beta_{10} y^{-\lambda/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (150)
\end{aligned}$$

$$\begin{aligned}
&= \sum_{k_1=0}^{\beta_5-1} \alpha_{81} \frac{\bar{\alpha}_r^{0.5} (-\lambda)^{\alpha_{83}-0.5} (\beta'_3)^{-\alpha_{83}}}{(2\pi)^{\frac{\bar{\alpha}_r - \lambda - 2}{2}}} \times \\
&\quad G_{-\lambda, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda} \left[\frac{\beta_{10}^{\bar{\alpha}_r} (\bar{\alpha}_r)^{-\bar{\alpha}_r}}{(\beta'_3)^{-\lambda} (-\lambda)^{\lambda}} \left| \begin{matrix} \Delta(-\lambda, 1 - \alpha_{83}) \\ \Delta(\bar{\alpha}_r) \end{matrix} \right. \right], \quad (151)
\end{aligned}$$

$$I_{39} = \alpha_{84} \int_0^\infty x^{\beta_{52}} e^{-\beta_{53} x^{-\lambda_{re}}} dx \quad (152)$$

$$= \alpha_{84} \frac{\Gamma(\nu_{11})}{n_{11} (\beta_{53})^{\nu_{11}}}, \quad (153)$$

$$I_{40} = \sum_{k_1=0}^{\delta_5-1} \alpha_{85} \int_0^\infty x^{\alpha_{86}} e^{-\delta'_3 x^{\bar{\alpha}_d} - \beta_{53} x^{-\lambda_{re}}} dx, \quad (154)$$

$$\begin{aligned}
&= \sum_{k_1=0}^{\delta_5-1} \alpha_{85} \int_0^\infty y^{\alpha_{87}-1} G_{0,1}^{1,0} \left[\delta'_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\
&\quad G_{0,1}^{1,0} \left[\beta_{53} y^{-\lambda/\bar{\alpha}_d} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (155)
\end{aligned}$$

$$= \sum_{k_1=0}^{\delta_5-1} \gamma_{83} G_{-\lambda_{re}, \bar{\alpha}_d}^{\bar{\alpha}_d, -\lambda_{re}} \left[\gamma_{84} \left| \begin{matrix} \Delta(-\lambda_{re}, 1 - \alpha_{87}) \\ \Delta(\bar{\alpha}_d) \end{matrix} \right. \right], \quad (156)$$

So,

$$S_1 = I_{37} - I_{38}, \quad (157)$$

$$S_2 = I_{39} - I_{40}, \quad (158)$$

where $\beta'_{55} = \beta_{55} (a_3)^{\beta_{54}}$, $\beta'_3 = \beta_3 (a_3)^{\bar{\alpha}_r}$, $\alpha_{80} = \beta_6 \beta_8$, $\alpha_{81} = \beta'_{55} \beta_8$, $\alpha_{82} = \beta_{12} + \beta_9$, $\alpha_{83} = \frac{\alpha_{82} + 1}{\bar{\alpha}_r}$, $\beta'_{13} = \beta_{13} (a_3)^{\beta_{14}}$, $\delta'_3 = \delta_3 (a_3)^{\bar{\alpha}_d}$, $\alpha_{84} = \delta_6 \beta_{51}$, $\alpha_{85} = \beta'_{13} \beta_{51}$, $\alpha_{86} = \beta_{14} + \beta_{52}$, $\alpha_{87} = \frac{\alpha_{86} + 1}{\bar{\alpha}_d}$, $m_{10} = \beta_9$, $n_{10} = -\lambda$, $\nu_{10} = \frac{m_{10} + 1}{n_{10}}$, $m_{11} = \beta_{52}$, $n_{11} = -\lambda_{re}$, $\nu_{11} = \frac{m_{11} + 1}{n_{11}}$, $\gamma_{83} = \alpha_{85} \frac{\bar{\alpha}_d^{0.5} (-\lambda_{re})^{\alpha_{87}-0.5} (\delta'_3)^{-\alpha_{87}}}{(2\pi)^{\frac{\bar{\alpha}_d - \lambda_{re} - 2}{2}}}$, $\gamma_{84} = \frac{\beta_{53}^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}{(\delta'_3)^{-\lambda_{re}} (-\lambda_{re})^{\lambda_{re}}}$.

2) *Strictly Positive Secrecy Capacity (SPSC) Equation for Scenario III:*

$$I_{44} = \alpha_{80} \int_0^\infty x^{\beta_9} e^{-\beta_{10} x^{-\lambda}} dx \quad (159)$$

$$= \alpha_{80} \frac{\Gamma(\nu_{10})}{n_{10} (\beta_{53})^{\nu_{10}}}, \quad (160)$$

$$I_{45} = \sum_{k_1=0}^{\beta_5-1} \alpha'_{81} \int_0^\infty x^{\alpha_{82}} e^{-\beta_3 x^{\bar{\alpha}_r} - \beta_{10} x^{-\lambda}} dx, \quad (161)$$

$$\begin{aligned}
&= \sum_{k_1=0}^{\beta_5-1} \alpha'_{81} \int_0^\infty y^{\alpha_{83}-1} G_{0,1}^{1,0} \left[\beta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times \\
&\quad G_{0,1}^{1,0} \left[\beta_{10} y^{-\lambda/\bar{\alpha}_r} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (162)
\end{aligned}$$

$$= \sum_{k_1=0}^{\beta_5-1} \alpha'_{81} \frac{\bar{\alpha}_r^{0.5} (-\lambda)^{\alpha_{83}-0.5} (\beta_3)^{-\alpha_{83}}}{(2\pi)^{\frac{\bar{\alpha}_r-\lambda-2}{2}}} \times G_{-\lambda, \bar{\alpha}_r}^{\bar{\alpha}_r, -\lambda} \left[\frac{\beta_{10}^{\bar{\alpha}_r} (\bar{\alpha}_r)^{-\bar{\alpha}_r}}{(\beta_3)^{-\lambda} (-\lambda)^{\lambda}} \left| \frac{\Delta(-\lambda, 1 - \alpha_{83})}{\Delta(\bar{\alpha}_r)} \right. \right], \quad (163)$$

$$I_{46} = \alpha_{84} \int_0^\infty x^{\beta_{52}} e^{-\beta_{53} x^{-\lambda_{re}}} dx \quad (164)$$

$$= \alpha_{84} \frac{\Gamma(\nu_{11})}{n_{11} (\beta_{53})^{\nu_{11}}}, \quad (165)$$

$$I_{47} = \sum_{k_1=0}^{\delta_5-1} \alpha'_{85} \int_0^\infty x^{\alpha_{86}} e^{-\delta_3 x^{\bar{\alpha}_d} - \beta_{53} x^{-\lambda_{re}}} dx, \quad (166)$$

$$= \sum_{k_1=0}^{\delta_5-1} \alpha'_{85} \int_0^\infty y^{\alpha_{87}-1} G_{0,1}^{1,0} \left[\delta_3 y \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] \times G_{0,1}^{1,0} \left[\beta_{53} y^{-\lambda/\bar{\alpha}_d} \left| \begin{matrix} - \\ 0 \end{matrix} \right. \right] dy, \quad (167)$$

$$= \sum_{k_1=0}^{\delta_5-1} \alpha'_{85} \frac{\bar{\alpha}_d^{0.5} (-\lambda_{re})^{\alpha_{87}-0.5} (\delta_3)^{-\alpha_{87}}}{(2\pi)^{\frac{\bar{\alpha}_d-\lambda_{re}-2}{2}}} \times G_{-\lambda_{re}, \bar{\alpha}_d}^{\bar{\alpha}_d, -\lambda_{re}} \left[\frac{\beta_{53}^{\bar{\alpha}_d} (\bar{\alpha}_d)^{-\bar{\alpha}_d}}{(\delta_3)^{-\lambda_{re}} (-\lambda_{re})^{\lambda_{re}}} \left| \frac{\Delta(-\lambda_{re}, 1 - \alpha_{87})}{\Delta(\bar{\alpha}_d)} \right. \right], \quad (168)$$

So,

$$\mu_1 = I_{44} - I_{45}, \quad (169)$$

$$\mu_2 = I_{46} - I_{47}, \quad (170)$$

where $\alpha_{80} = \beta_6 \beta_8$, $\alpha_{81} = \beta_5 \beta_8$, $\alpha_{82} = \beta_{12} + \beta_9$, $\alpha_{83} = \frac{\alpha_{82}+1}{\bar{\alpha}_r}$, $\alpha_{84} = \delta_6 \beta_{51}$, $\alpha_{85} = \beta_{13} \beta_{51}$, $\alpha_{86} = \beta_{14} + \beta_{52}$, $\alpha_{87} = \frac{\alpha_{86}+1}{\bar{\alpha}_d}$, $m_{10} = \beta_9$, $n_{10} = -\lambda$, $\nu_{10} = \frac{m_{10}+1}{n_{10}}$, $m_{11} = \beta_{52}$, $n_{11} = -\lambda_{re}$, $\nu_{11} = \frac{m_{11}+1}{n_{11}}$

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